

Dynamic Functional Graph Learning for Subject-Independent Consumer Engagement Decoding from EEG and Eye Tracking: A Leakage-Aware NeuMa Study

Priyadharshini D^[0009-0004-2291-1814]¹ and Shridevi S^[0000-0002-0038-7212]^{1*}

^{1*}Centre for Neuroinformatics, Vellore Institute of Technology, Chennai, India.

*Corresponding author(s). E-mail(s): shridevi.s@vit.ac.in;
Contributing authors: priyadharshini.2024b@vitstudent.ac.in;

Abstract

Consumer engagement drives advertising effectiveness, yet self-reports are retrospective and biased. Electroencephalography (EEG) and eye tracking offer objective alternatives, but multimodal decoders are rarely evaluated subject-independently or tested for whether components do as claimed. We present the Interpretable Neuro-Symbolic Hybrid Dynamic Graph Spiking Cross-Modal Transformer (INS-HDGS-CMT), which encodes EEG epochs as dynamic functional-connectivity graphs with graph-attention and spiking modules and fuses EEG and eye tracking in a cross-modal transformer with a soft-rule inspection module. We evaluate it on the NeuMa dataset under leave-one-subject-out cross-validation, with controls locating where accuracy arises. Because the NeuMa label is a fixed rule over EEG band power and gaze statistics, we quantify its linear recoverability: area under the receiver operating characteristic curve (ROC-AUC) 0.67 from EEG terms, 0.92 from gaze. The full model reaches ROC-AUC 0.88 and balanced accuracy 0.75, matching, not exceeding, the best tuned eye-tracking and fusion baselines (Holm-corrected). A gaze-free EEG branch scores 0.59, indistinguishable from eight tuned EEG encoders. Removing the graph pathway costs 0.09 balanced accuracy, yet density-matched static or random graphs cost nothing, so it acts through node-feature processing, not measured connectivity. The learned gate bypasses the rules; closing it retains balanced accuracy 0.72 and ROC-AUC 0.84, yielding electrode- and band-level rule groundings. Neither modality predicts purchase within 5-s epochs (ROC-AUC 0.61), whereas session-level dwell-time probes do (0.85–0.90). Subject-independent decoding of the rule-defined index is achievable but gaze-limited, the graph and spiking components are representations, not accuracy sources, and short-window engagement claims need behavioural validation this dataset lacks.

Keywords: Consumer engagement, Neuromarketing, Electroencephalography, Eye tracking, Graph neural networks, Multimodal learning, Dynamic functional connectivity

1 Introduction

Consumer engagement, the degree to which a consumer attends to, processes, and is emotionally moved by a marketing stimulus, is closely linked to advertising effectiveness, brand recall, and purchase intention [1, 2]. It has conventionally been measured by self-report, but such

instruments are retrospective and introspective, presuppose that consumers can verbalise covert cognitive and affective states, and are susceptible to social-desirability, demand, and recall biases [2, 3]. Much of purchasing behaviour is instead driven by rapid, largely unconscious attentional and emotional processes on sub-second

timescales that elude verbal report [3], motivating consumer neuroscience and, increasingly, artificial-intelligence-based analysis in the search for objective, time-resolved correlates of engagement [1, 4, 5]. Among neurophysiological modalities, electroencephalography (EEG) is dominant in neuromarketing for its millisecond resolution, portability, low cost, and non-invasiveness [1, 6, 7]; engagement is expressed through oscillatory dynamics such as frontal-theta and parietal-alpha (de)synchronisation and event-related components [1, 8], which deep-learning models increasingly recover [9, 10]. EEG does not, however, reveal where a consumer is looking; eye tracking (ET) supplies exactly this, indexing overt visual attention through fixations, dwell time, saccades, scan-paths, and pupillary responses [3, 8, 11]. The two modalities are naturally complementary, EEG capturing cognition without gaze and ET capturing gaze without cognition [8], which is the central rationale for multimodal EEG–ET decoding of consumer engagement.

An expanding literature pairs EEG with ET, or enriches EEG with attention and connectivity features, for consumer-relevant classification [8, 12–17]; most directly, hybrid methods on the NeuMa data combine EEG graph signal processing with gaze dynamics [18], and graph neural networks are now a prominent approach to EEG decoding [19–22], alongside spatial-attention and cross-modal EEG–ET fusion architectures [12, 23–26]. In parallel, EEG decoding architectures have advanced rapidly: convolutional attention networks that focus jointly on temporal and spectral structure, such as BCINetV1 [56], exemplify recent convolutional-attention approaches to motor-imagery decoding, and large pretrained EEG transformers, notably LaBraM [54], trained on some 2,500 hours of heterogeneous recordings with a neural tokenizer, and EEGPT [55], a masked-autoencoding transformer yielding universal channel-wise representations, provide generic representations that transfer across datasets and tasks; adaptive tokenization schemes that segment EEG by its intrinsic dynamics and route tokens to band-specific codebooks have further improved the interpretability of such embeddings [57]. These models are single-modality, are pretrained on corpora far larger than any neuromarketing dataset, and are evaluated on

motor-imagery, abnormality, event-type or emotion targets; their use as drop-in EEG encoders within a multimodal engagement framework, and their behaviour under subject-held-out evaluation on a 42-participant cohort, remain open questions that the present study does not address directly but that its leakage-aware protocol is designed to make answerable (Section 4.6). Four limitations nonetheless recur. First, subject-mixed train/test splits leak subject-specific idiosyncrasies and overstate generalisation [13, 27–31]. Second, fusion by plain concatenation or late averaging captures neither the conditional dependence between gaze and neural activity nor the differing reliability of the two streams [23]. Third, EEG functional connectivity is usually treated as static within a trial, discarding its non-stationary, dynamically reconfiguring character. Fourth, the resulting models are typically opaque, lacking the interpretable, domain-consistent explanations that practitioners and ethics boards require for trustworthy deployment [32, 33]. These delineate a gap: a multimodal EEG–ET framework validated under a genuinely subject-independent protocol, in which the architectural responses to the other three limitations, a dynamic connectivity graph, learned cross-modal attention and an inspectable soft-rule module, are not assumed to help but tested. The proposed framework, the Interpretable Neuro-Symbolic Hybrid Dynamic Graph Spiking Cross-Modal Transformer (INS-HDGS-CMT), instantiates those responses and is evaluated on the NeuMa dataset [34] under leave-one-subject-out cross-validation (LOSOCV), a direction the prior graph-based NeuMa study itself identified as promising [18], together with the controls needed to say which of its components the accuracy depends on.

A natural objection is that eye tracking already provides a comparatively direct index of overt visual attention, so why decode engagement from EEG at all, let alone train an EEG model alongside eye-tracking information? Three considerations motivate the design. First, gaze indexes *overt* attention, where the eyes point, whereas engagement as defined in consumer neuroscience also comprises covert processing, arousal and affective evaluation that need not be expressed in the scanpath [1, 8]; because the NeuMa engagement label is itself constructed from gaze statistics

(Section 2.4), an eye-tracking classifier that reproduces it demonstrates only that the label is recoverable from its own source, whereas an EEG classifier that recovers it beyond what the label’s own EEG terms yield would show that the neural signal carries engagement-related information independent of the label’s gaze construction. This is the question the gaze-free EEG branch is built to answer; Section 3.2 reports that on this label the answer is negative. Second, the two modalities fail in different situations: eye tracking requires a calibrated tracker and stimulus-locked screen coordinates, is uninformative for audio-only advertising formats where there is no visual target to fixate, and, for the fixed-grid region-of-interest (ROI) representation used here, is most directly applicable to static or slowly-changing layouts rather than dynamic video, whereas EEG is portable, stimulus-agnostic and unaffected by the visual layout of the stimulus [6, 7]. A framework whose EEG pathway is evaluated on its own, and whose multimodal pathway quantifies what gaze adds when it is available, therefore covers both deployment settings. Third, within the multimodal model, gaze is not merely a second input: the cross-modal contrastive objective (Section 2.5.9) uses the concurrent gaze embedding as a training-time signal for the EEG embedding, and the ROI vector gates the EEG graph, which are two mechanisms by which eye-tracking information could shape the neural pathway. Whether it does is not assumed: the contribution of the gaze pathway is quantified by paired comparison of the full model with variants that remove the gaze sequence, the ROI vector, or both (Section 3.5), and the result is that the gaze inputs carry the accuracy rather than improve an EEG core.

INS-HDGS-CMT addresses these gaps through six contributions, integrated within a single end-to-end framework and evaluated under leakage-aware, subject-independent protocols:

- **Dynamic functional-graph modelling of EEG, with a direct test of what it contributes.** Each epoch is represented as a sequence of functional-connectivity graphs (Pearson correlation; Eq. (2)) whose edges evolve across successive sub-windows, rather than the static per-trial connectivity matrix typical of consumer-EEG models. Removing the

graph pathway costs 0.09 in balanced accuracy, but replacing the measured adjacency by density-matched static or random graphs costs nothing (Section 3.4): the pathway acts through its node-feature processing, not through the measured connectivity, and the gaze-free branch built on it is statistically indistinguishable from eight individually tuned EEG encoders (Section 3.2).

- **Cross-modal EEG–eye-tracking fusion.** Rather than concatenating or late-averaging modality features, a cross-modal transformer lets the EEG representation query both the connectivity-graph and eye-tracking context. Against eighteen individually tuned baselines it reaches the same operating-point accuracy and ranking quality as the best eye-tracking-only and fusion models (no difference survives Holm correction; Section 3.2), and the pathway analysis of Section 3.5 shows that the gaze inputs carry the accuracy.
- **Spiking temporal encoding.** A leaky integrate-and-fire (LIF) front-end provides an event-driven EEG representation. Ablation leaves accuracy statistically unchanged, so the encoder is not claimed to be predictively necessary; it is retained as an event-driven representation (trained firing rate 7.5–10.6% on the checkpoints measured, about 90% spike sparsity) for which no energy saving is claimed: on the central processing unit (CPU) and graphics processing unit (GPU) used here it is measurably slower than a dense twin, and on a positive control (resting state versus browsing) the encoder trained alone decodes at an area under the receiver operating characteristic curve (ROC-AUC) of 0.55 where convolutional encoders reach 0.92–0.94 (Section 3.4).
- **A differentiable soft-rule module for inspecting rule-activation patterns.** An attention-weighted mixture of soft rules exposes per-sample activation traces. The fidelity of the rule pathway to the final decision is measured directly, through the learned bypass gate, the agreement between rule-only and gated decisions, and a variant trained with the gate closed (Section 3.11); the rule premises are grounded in scalp electrodes and frequency bands by integrated-gradient projection rather than reported as latent indices. The test shows that the gated model bypasses its rules; faithful

rule explanations are obtained from the model trained with the gate closed, at a cost of two to three points of balanced accuracy, and are reported from that model.

- **Explainability grounded in hypothesis testing.** A within-subject single-marker concordance analysis shows each canonical univariate marker is individually non-significant, whereas the same epochs are separated at ROC-AUC 0.88 by the full model and only 0.59 by the gaze-free branch (Section 3.7), indicating that the index is carried by its gaze terms and that whatever the EEG contributes is distributed across electrodes and bands rather than concentrated in any isolated biomarker.
- **Leakage-aware, control-driven subject-independent evaluation.** Evaluation is performed under subject-held-out LOSOCV, with the label’s construction from EEG band power and gaze statistics disclosed and its recoverability from each modality quantified, all baselines individually tuned under the same nested validation protocol, a gaze-free EEG branch, a retraining under fold-wise labels, density-matched graph-topology nulls, a rule-fidelity test, a positive control on the same recordings and a behavioural purchase label (Sections 3.2–3.6). Under these controls the multimodal accuracy is gaze-limited and the physiology within a 5-s epoch does not predict purchase, which we report as the study’s principal finding.

These contributions are of unequal kind. The first four are architectural, and under the controls of the sixth none of them turns out to be the source of the accuracy; the evaluation is the contribution we consider durable, and Section 4.2 sets out, component by component, where the accuracy comes from.

The principal contribution of this work does not lie in the individual use of graph neural networks, transformers, or spiking neural networks, each of which has previously been investigated in related domains. Instead, the contribution arises from subjecting a representative multimodal neuromarketing architecture to leakage-aware, subject-independent evaluation with the controls needed to say where its accuracy comes from, and from reporting the outcome in full, including the claims of the submitted version that the controls overturn.

The remainder of the paper is organised as follows. Section 2 describes the NeuMa dataset, the preprocessing and synchronisation, the INS-HDGS-CMT framework, and the evaluation protocol. Section 3 presents the experimental results, including the explainability analysis and the high/low-engagement case studies; Section 4 interprets the findings; and Section 5 concludes.

2 Materials and Methods

The contributions above are realised through a single end-to-end framework, the Interpretable Neuro-Symbolic Hybrid Dynamic Graph Spiking Cross-Modal Transformer (INS-HDGS-CMT), for subject-independent consumer engagement prediction from EEG and eye tracking. This section describes, in turn, the NeuMa dataset and participants, the modality-specific preprocessing and EEG–ET synchronisation, the components of the proposed architecture, and the leave-one-subject-out evaluation protocol under which all models are assessed.

2.1 NeuMa Dataset and Participants

The publicly described NeuMa dataset of synchronously acquired EEG and eye-tracking recordings, collected while participants viewed marketing stimuli, was analysed [34]. The public release contains 44 participants; the $N = 42$ participants for whom per-subject engagement labels are available were retained (two participants without such labels were excluded), with the sex and age distributions as reported in the original dataset publication [34].

EEG was recorded with a Wearable Sensing DSI-24 headset at 300 Hz. Of its 24 physical channels, 19 correspond to scalp EEG electrodes positioned according to the international 10–20 system (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz), whereas the remaining channels do not carry cortical signals: three auxiliary/electrooculography (EOG) channels, one mastoid reference channel (A2, recorded separately for optional re-referencing; the primary reference, A1, is applied in hardware and is not itself a digitised channel), and one trigger channel. In line with all prior analyses of this dataset, those of the dataset authors [18] and subsequent

NeuMa engagement paradigm: real ROI, gaze and EEG for one recorded epoch

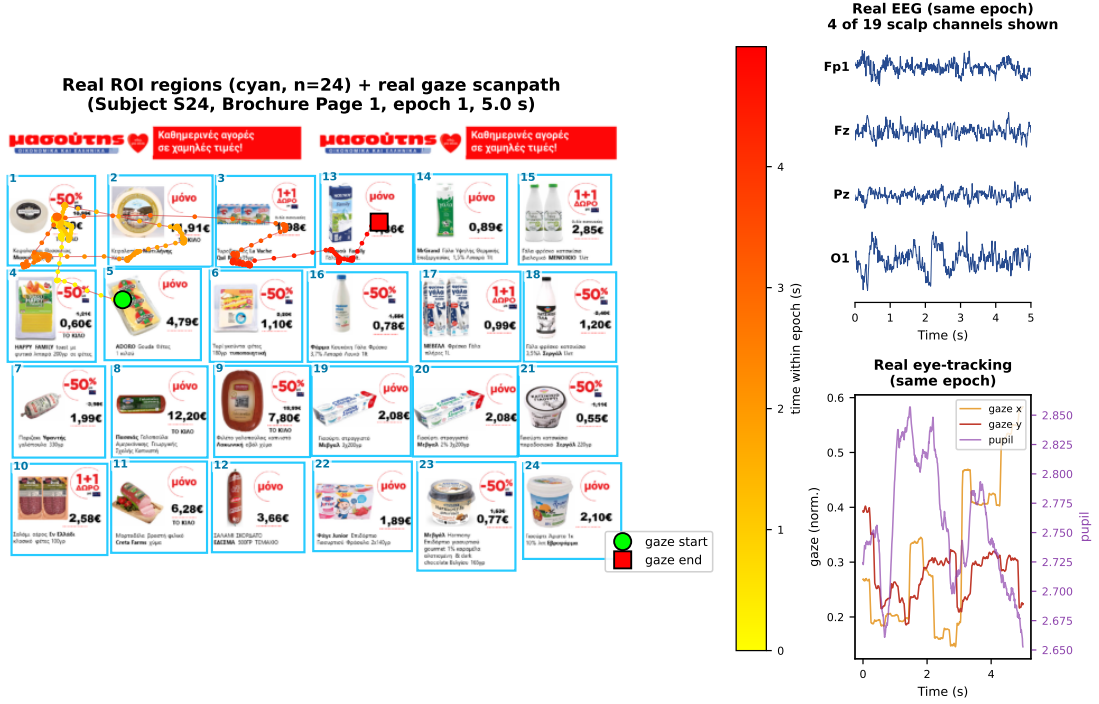


Fig. 1 A synchronised multimodal epoch (subject S24): the brochure page with the gaze scanpath overlaid (left), a subset of the retained scalp-EEG channels (top right) and the concurrent gaze-position and pupil-diameter traces (bottom right), all within the same 5 s epoch.

multimodal and graph-neural-network studies [35] alike, the 19 scalp electrodes are treated as graph nodes and the reference and auxiliary channels discarded, as the latter would otherwise inject non-neural (in particular trigger-related) information into the connectivity graph. This selection is enforced in the dataset loader, and every result reported here comes from runs under it. (The run reported in the submitted version had been trained before the selection was enforced and carried all 24 recorded channels as nodes; it has been replaced throughout by the re-run of the same configuration under the 19-electrode montage, which reproduces it within 0.01 in balanced accuracy and 0.02 in ROC-AUC.) During each session a participant viewed a structured sequence of marketing stimuli, brochure pages partitioned into a fixed grid of regions of interest (ROIs) corresponding to individual product placements, while EEG and ET were recorded concurrently. A representative recorded epoch is shown in Fig. 1

for subject S24, the participant examined as the HIGH-engagement exemplar in the case study of Section 3.12: the ROI grid defined over the stimulus, the corresponding gaze scanpath, a subset of the retained EEG channels, and the concurrent gaze-position and pupil-diameter traces jointly illustrate the synchronised, multimodal recording that underlies both the dynamic-graph EEG encoding (Section 2.5.2) and the engagement labelling rule (Supplementary Table S5). The dataset characteristics used here are listed in Table 1.

2.2 EEG Preprocessing

EEG signals were re-referenced to the common average, band-pass filtered to 1–45 Hz, and notch filtered at 50 Hz to suppress line noise. Ocular and muscular artefacts were attenuated with independent component analysis (15 components); for one participant (S01) four non-cortical channels flagged as noisy were zero-filled, and no scalp

Table 1 NeuMa dataset characteristics.

Characteristic	Value
Subjects	42 analysable of 44 released (37 evaluable under global labels; 42 under subject-median)
EEG channels	19 (10–20 scalp)
EEG sampling rate	300 Hz
ET features	3 (gaze x , gaze y , pupil)
ET sampling rate	120 Hz
Epoch length	5.0 s
Total epochs	385
Class distribution (HIGH:LOW)	193:192

electrode was removed for any participant. The signals were kept at their native 300 Hz sampling rate and segmented into fixed-length 5 s epochs (Section 2.4). Per-channel z -score normalisation was performed within each subject, using that subject’s own statistics computed without reference to the labels. Because no statistics cross subject boundaries, the held-out test subject is normalised solely from its own (unlabelled) data; this procedure is label-free but transductive per-subject normalisation under LOSOCV, mirroring the per-user normalisation that a deployed system would apply to a new participant. Because it nonetheless uses statistics of the whole held-out recording, the full model was also retrained in the revision with per-epoch instance normalisation, which uses no subject-level statistic at all; it reached the same accuracy (Section 3.8).

2.3 Eye-Tracking Preprocessing

Raw gaze coordinates and pupil diameters were cleaned by discarding samples lost to blinks and tracking dropouts, with short gaps filled by linear interpolation. Gaze positions were used to derive behavioural attention indices, namely fixations, fixation durations, and dwell time over areas of interest (Section 2.5.3). Guided by a feature-ablation analysis, a compact, un-normalised three-channel eye-tracking representation (gaze x , gaze y , pupil) was retained, since higher-dimensional configurations did not improve, and in some cases degraded, subject-independent performance.

2.4 EEG–Eye Tracking Synchronisation and Epoch Generation

The two modalities were temporally aligned using shared stimulus-onset markers and timestamps, with each stream kept at its native sampling rate (EEG 300 Hz, ET 120 Hz) rather than resampled to a common base. The synchronised recordings were segmented into non-overlapping 5.0 s epochs, each carrying paired EEG and ET tensors. Fixed-length windows were chosen over the gaze-defined, variable-duration epochs of prior NeuMa work [18] for two reasons. First, fixed windows produce equal-shape EEG and ET tensors, a requirement for batched dynamic-graph construction and transformer encoding without length-dependent padding. Second, because the engagement labels are derived in part from gaze statistics (Supplementary Table S5), gaze-defined epoch durations would tie epoch length to the label, whereas a fixed 5 s window removes this confound. The 5 s span also accommodates the sequence of sub-windows from which the dynamic functional graphs are built, leaving enough samples per window for connectivity estimation while preserving within-trial temporal resolution.

Engagement labels were recomputed directly on the new fixed 5 s epochs rather than inherited from the original variable-length product epochs, using the multimodal labelling rule of the dataset pipeline (`engagement_phase3d.py`; Supplementary Table S5). For each 5 s stimulus epoch, five EEG terms are computed from the mean of the first four channels of the recording montage, P3, C3, F3 and Fz (Welch band power of that mean signal in theta 4–8 Hz, alpha 8–13 Hz and beta 13–30 Hz, the theta/beta ratio,

and an ‘asymmetry’ term that is the difference in mean amplitude between P3 and Fz). The pipeline names these channels ‘frontal’, and the submitted version repeated that name; two of the four are in fact parietal and central electrodes, and the asymmetry term, a time-domain mean difference of band-passed, average-referenced signals, is close to zero by construction and contributes little. The five gaze terms are computed from the epoch’s own gaze sequence (mean absolute horizontal and vertical gaze position, horizontal gaze dispersion, the number of consecutive samples whose horizontal displacement is below 0.05 of the screen width, and the Shannon entropy of a 20-bin horizontal gaze histogram). All ten terms are min–max normalised across the pooled epochs of all subjects and combined with fixed weights (theta +1.5, beta +1.2, alpha –1.5, theta/beta ratio +1.0, asymmetry +0.5, the four positive gaze terms +1.0 each, gaze entropy –1.0); the composite score is binarised at the pooled median (global regime; 385 epochs, 193 HIGH and 192 LOW) or at each subject’s own median (subject-median regime, Section 3.10). Pupil diameter does not enter the label. Because the scaler and the median are fitted on the pooled epochs, each held-out subject contributes to the definition of its own labels; refitting both without the held-out subject (leave-one-subject-out) changes 9 of the 385 labels (2.3%, five subjects), and re-scoring the saved held-out probabilities of the full model against those labels moves ROC-AUC from 0.879 to 0.871 and leaves balanced accuracy at 0.747 and Matthews correlation coefficient (MCC) at 0.48 (`scripts/analysis/label_fold_sensitivity.py`), within the run-to-run variation of the model (Section 3.4); retraining the model under the fold-wise label, rather than re-scoring its predictions, gives the same conclusion (Section 3.10, Supplementary Table S17). The label is thus a fixed, rule-defined multimodal index rather than a behavioural criterion; because both EEG band power and gaze statistics enter it, its recoverability from each modality’s defining features is quantified explicitly (Section 3.2, Supplementary Table S15). No labels are aggregated or otherwise transferred from the original product-epoch boundaries.

Each epoch received a binary consumer engagement label (HIGH vs. LOW), and two complementary labelling regimes are considered. The primary regime applies a single global engagement

threshold consistently across subjects, giving an absolute, cross-subject-coherent notion of engagement; under it, five subjects (S16, S31, S33, S41, and S44) carry only one class and are therefore excluded from the test folds (37 of 42 subjects evaluable) while remaining in every training fold. As a complementary coverage regime, each subject is additionally split at their own rank-based median, which guarantees both classes for every participant and renders all 42 subjects evaluable; this captures within-subject relative engagement rather than an absolute level. The global threshold is treated as the headline definition, an unconstrained per-subject threshold defines engagement only relative to each individual and destabilises cross-subject labelling, and the all-42 subject-median evaluation is reported as a robustness and subject-coverage analysis in Section 3.10.

2.5 Proposed INS-HDGS-CMT Framework

2.5.1 Overview of the Proposed Framework

INS-HDGS-CMT is an end-to-end multimodal architecture of six interacting components, shown in Fig. 2. The EEG stream is converted into a sequence of dynamic functional graphs, encoded by a spiking neural front-end, and then processed by hierarchical dynamic graph attention and a temporal transformer. In parallel, the ET stream is encoded into behavioural attention representations. A cross-modal transformer integrates the two representations, and a neuro-symbolic reasoning module refines the fused representation before the final engagement prediction. The pipeline is summarised stage by stage in Table 2, implementation details are listed in Table 3, and a summary of the mathematical notation used throughout this section is given in Supplementary Table S4. Five loss terms shape training—focal class-balanced classification, a cross-modal EEG–eye-tracking contrastive term, rule regularisation, and two subject-alignment terms (maximum mean discrepancy, MMD, and domain-adversarial)—as described in Section 2.5.9; an ROI-guidance term was implemented but is inactive on the training path and is not part of the effective objective. The dynamic functional graphs are precomputed per window and are not updated by gradients, so

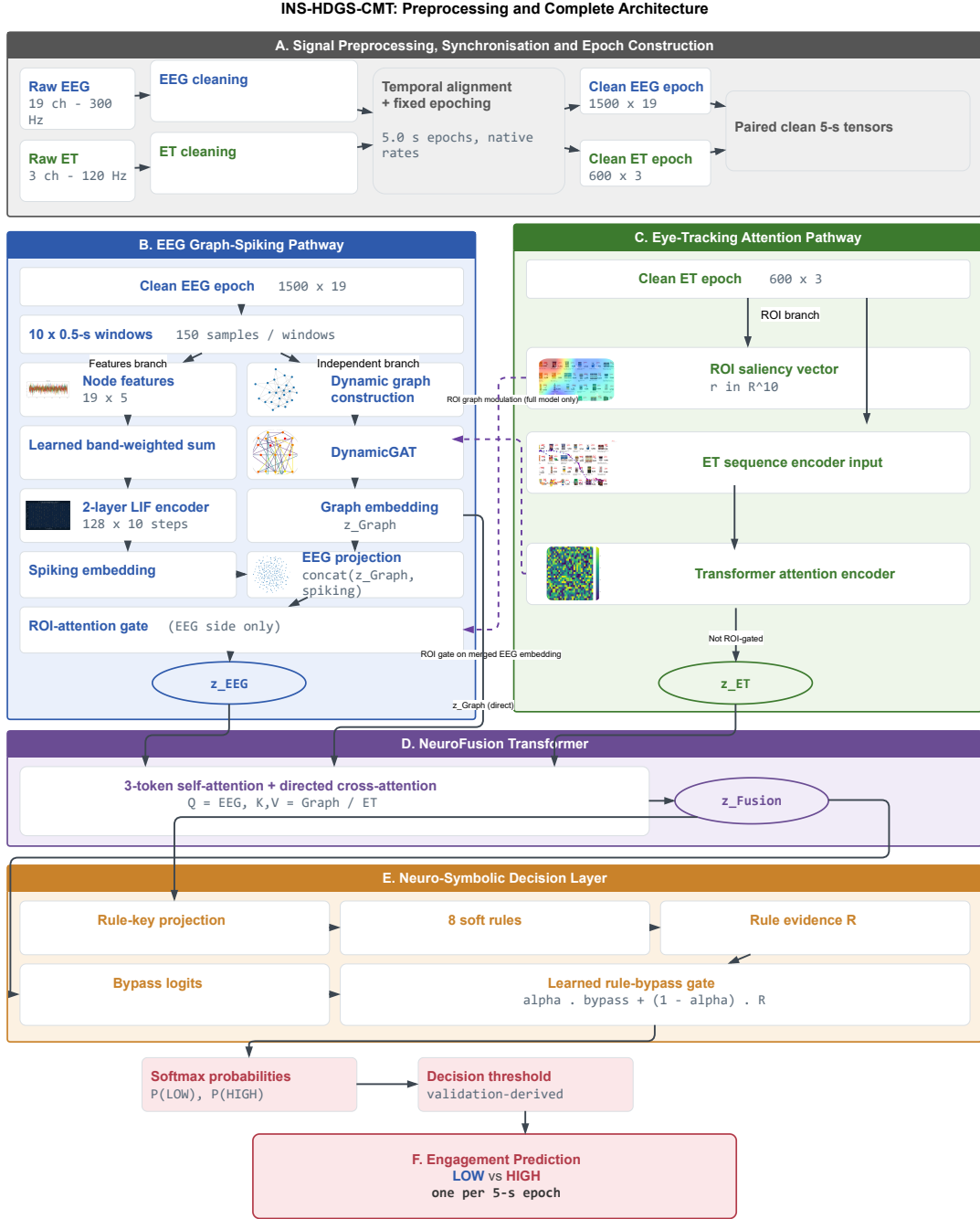


Fig. 2 Overview of INS-HDGS-CMT. Signal preprocessing, synchronisation and epoch construction (A) yield paired EEG and eye-tracking tensors that feed the EEG graph-spiking pathway (B) and the eye-tracking attention pathway (C); these are fused by the NeuroFusion Transformer (D), and the neuro-symbolic decision layer (E) yields the engagement prediction (F), one per 5-s epoch.

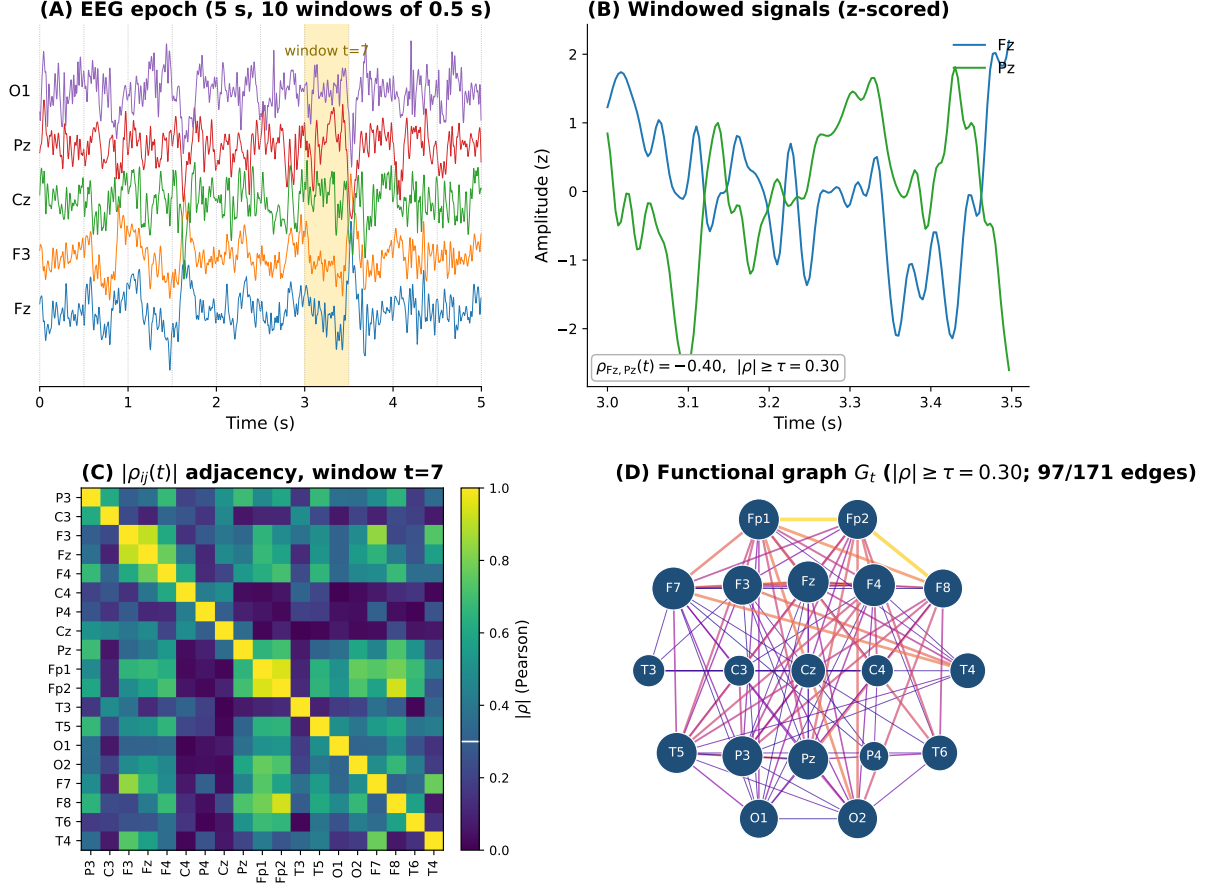


Fig. 3 Dynamic functional-graph construction from one real epoch (Pearson connectivity, Eq. (2)). (A) Ten 0.5 s windows of a 5 s epoch; the shaded window is shown in B–D. (B) Two z-scored electrode signals, whose correlation gives one edge weight. (C) The connectivity matrix $|C_t|$ of the window. (D) The graph G_t after thresholding at $\tau = 0.30$; node size scales with degree.

no graph-regularisation term enters the effective training objective. The ROI-modulation pathway shown with a dashed arrow is active only in this full multimodal configuration; the gaze-free EEG branch of Table 4 disables it entirely and receives no eye-tracking-derived input. Each component below is introduced with the purpose it was designed for and the section in which that purpose is tested; the outcomes of those tests, several of which withdraw the purpose, are stated in the pipeline table so that the Methods do not describe the architecture as more than the Results support. The construction of the dynamic functional graph is illustrated in Fig. 3; the cross-modal fusion and neuro-symbolic reasoning stages are shown in overview here and detailed separately in Fig. 4. A step-by-step summary of the pipeline, with the

operation and rationale of each stage, is given in Table 2.

2.5.2 Dynamic EEG Graph Construction

Each EEG epoch is modelled as a time-evolving functional graph rather than a static network. For a window indexed by t , the graph is

$$G_t = (V, E_t, A_t), \quad (1)$$

where V is the fixed set of EEG electrodes (nodes), E_t is the set of functional connections active in window t , and $A_t \in \{0, 1\}^{|V| \times |V|}$ is the binary adjacency matrix of that window, obtained by thresholding its weighted connectivity matrix

Table 2 Processing pipeline: operation, design purpose and the finding of Section 3 for each of the eight stages.

#	Stage (Section)	Operation	Designed purpose and test outcome
1	Dynamic EEG graph construction (Sec. 2.5.2)	Per-window Pearson correlation between electrode time-domain signals $\rightarrow$ sequence of graphs $\{G_t\}$.	Time-varying connectivity rather than a static matrix. Removing the pathway costs 0.09 balanced accuracy; replacing the measured adjacency by density-matched static or random graphs costs nothing (Sec. 3.4).
2	Eye-tracking feature encoding (Sec. 2.5.3)	Gaze- x , gaze- y , pupil streams $\rightarrow$ attention tokens.	Overt visual attention. Carries the accuracy on this label: removing all gaze input costs 0.29 ROC-AUC (Sec. 3.5).
3	Spiking encoding, LIF (Sec. 2.5.4)	LIF front-end over the ten per-window feature vectors $\rightarrow$ spike-based embedding, merged with the graph embedding.	Event-driven representation. No effect on accuracy (Sec. 3.4); slower than a dense twin on the CPU and GPU used; at chance on its own on the positive control (Supplementary Table S16).
4	Hierarchical graph attention (Sec. 2.5.5)	Attention weights electrodes and edges across the graph sequence.	Weights electrodes and edges per window. The coefficients are descriptive (Sec. 3.11); the topology nulls show the decision does not depend on them.
5	Temporal transformer (Sec. 2.5.6)	Self-attention over the sub-window sequence.	Models temporal evolution of connectivity.
6	Cross-modal fusion (Sec. 2.5.7)	Self-attention over the EEG, graph, and E/T tokens (mutual), then directed EEG-to-context cross-attention; gated residual fusion.	Lets the EEG representation query gaze context. No significant difference from simpler fusion designs or from a gaze-only encoder after correction (Sec. 3.2, Supplementary Table S10).
7	Neuro-symbolic decision layer (Sec. 2.5.8)	Fuzzy match to eight rule premises; rules vote HIGH/LOW; constraint aggregation $\rightarrow$ logits.	Inspectable rule-activation traces. The learned gate bypasses the rules in the production model; faithful traces come from the variant trained with the gate closed, at a cost of two to three points (Sec. 3.11).
8	Engagement prediction	Logits $\rightarrow P(\text{HIGH})$; decision at the operating threshold transferred from the validation subject, with the temperature-scaled 0.5 operating point reported separately.	Subject-independent output; threshold and temperature set on a validation subject, never on the test subject (Sec. 2.6).

$C_t \in \mathbb{R}^{|V| \times |V|}$ (Eq. (4)). The node set is therefore anatomically fixed while the edges reconfigure over time, letting the model follow the non-stationary reorganisation of cortical functional networks during stimulus viewing.

This differs from related graph-based EEG approaches, which typically construct a single functional-connectivity graph per trial or epoch and apply attention or multi-scale convolution over that fixed topology [19–22]. Here the connectivity topology itself, not only the node features, is treated as time-varying: a sequence of graphs $\{G_t\}_{t=1}^T$ is built per epoch, one per sub-window, and their evolution is subsequently modelled through hierarchical graph attention (Section 2.5.5) and a temporal transformer (Section 2.5.6).

Edge weights are derived from windowed functional connectivity between electrode pairs. The framework is agnostic to the specific measure populating C_t ; the reported results use the Pearson correlation coefficient between the windowed time-domain EEG signals of each electrode pair, computed independently per window t to capture

time-varying coupling:

$$\rho_{ij}(t) = \frac{\text{Cov}(x_i(t), x_j(t))}{\sigma_{x_i(t)} \sigma_{x_j(t)}}, \quad (2)$$

where $x_i(t)$ and $x_j(t)$ denote the time-domain EEG samples of electrodes i and j within window t ; each 5 s epoch is divided into ten non-overlapping 0.5 s windows (150 samples at 300 Hz). Relative band power across five canonical bands (each band’s share of the window’s total power, layer-normalised per window) enters the graph as electrode-node features, whereas $C_t(i, j) = \rho_{ij}(t)$ holds this window-specific amplitude correlation. Phase-locking value (PLV) [36], a phase-synchrony-based alternative particularly suited to narrowband oscillatory coupling, is

$$\text{PLV}_{ij} = \left| \frac{1}{N} \sum_{k=1}^N e^{j(\phi_i(k) - \phi_j(k))} \right|, \quad (3)$$

where N is the number of samples in the window, $\phi_i(k)$ and $\phi_j(k)$ are the instantaneous phases of electrodes i and j at sample k (obtained via

the Hilbert transform), and j is the imaginary unit. $PLV_{ij} \in [0, 1]$ quantifies the consistency of the phase difference between the two electrodes: values near unity denote stable phase coupling, a widely used marker of functional connectivity that is comparatively insensitive to amplitude differences across channels and subjects. PLV was evaluated as an alternative connectivity measure (Section 3.10); Pearson correlation was retained as the reported measure for its lower computational cost and greater numerical stability at the small per-subject sample sizes inherent to LOSOCV, in which each held-out subject contributes about nine test epochs.

To obtain a sparse, interpretable graph, connectivity is thresholded:

$$A_t(i, j) = \begin{cases} 1, & |C_t(i, j)| \geq \tau \text{ or } i = j, \\ 0, & \text{otherwise,} \end{cases} \quad (4)$$

where $C_t(i, j)$ is the chosen connectivity measure ($\rho_{ij}(t)$ of Eq. (2) for the reported results) and τ is a connectivity threshold; the graph attention of Section 2.5.5 uses A_t as a binary neighbourhood mask, so the connectivity magnitude determines whether an edge exists but not the weight of the message passed along it, which is learned by the attention coefficients. Thresholding sparsifies the adjacency prior to attention-based aggregation, a common step in graph-based EEG connectivity modelling used to suppress spurious weak couplings and control the effective graph density [19–22], and produces a sparse adjacency. The specific value ($\tau = 0.30$, Table 3) was fixed a priori and held constant across all participants and folds; it was not tuned per fold. Its consequences are quantified in two ways (Section 3.10, Supplementary Table S7 and Fig. S4). Structurally, over all 3,850 windows of the 385 epochs, $\tau = 0.30$ retains $57 \pm 15\%$ of the 171 electrode pairs (mean degree 10.3), on the smooth part of the density curve: density falls almost linearly from 0.85 at $\tau = 0.10$ to 0.15 at $\tau = 0.70$, whereas graph fragmentation, measured as electrodes left with fewer than three edges, is negligible up to $\tau = 0.30$ (0.5 isolated electrodes per window) and rises steeply beyond $\tau = 0.40$ (2.0) to $\tau = 0.50$ (4.9). Predictively, the gaze-free EEG branch was re-trained under the full 37-subject LOSOCV protocol for $\tau \in \{0.20, 0.25, 0.30, 0.35, 0.40, 0.50\}$

(Section 3.10). Whether the measured topology matters at all, at any τ , is a separate question and is tested directly by replacing every window’s adjacency with a density-matched static or random graph (Section 3.4). The construction of the dynamic functional graph is depicted in Fig. 3 for one 0.5 s window of a real epoch, using the Pearson correlation of Eq. (2) exactly as the model computes it: the windowed signals of an electrode pair (panel B) give one entry of the $|\rho_{ij}(t)|$ adjacency (panel C), which is thresholded at $\tau = 0.30$ to form G_t (panel D).

2.5.3 Eye-Tracking Behavioural Feature Representation

The ET stream is summarised by behavioural attention features computed from gaze and pupil signals. For a fixation spanning onset t_{onset} to offset t_{offset} , the fixation duration is

$$FD = t_{\text{offset}} - t_{\text{onset}}, \quad (5)$$

and the dwell time over a region or epoch is the sum of its constituent fixation durations,

$$DT = \sum_i FD_i. \quad (6)$$

Behaviourally, longer fixation durations and greater dwell time reflect sustained overt attention and deeper processing of a visual element, whereas short, dispersed fixations signal scanning or disengagement. These indices are used in the descriptive and case-study analyses of Sections 3.11 and 3.12; the ET encoder itself consumes the raw three-channel sequence of Section 2.3 (gaze x , gaze y , pupil; 600 samples per epoch), not hand-crafted indices. In addition, a region-of-interest (ROI) saliency vector $r \in \mathbb{R}^{10}$ is computed per epoch as the normalised count of gaze samples falling in each cell of a fixed 5×2 grid of the stimulus page (five columns by two rows); it is therefore a gaze-occupancy histogram, not a fixation-based dwell time, and is referred to as the ROI saliency vector throughout; r feeds the ROI-attention gating and ROI graph modulation of the full model and the ROI-guidance loss of Section 2.5.9. The 24 product boxes drawn in Fig. 1 are the brochure’s layout and are not a model input; the sensitivity of the saliency vector, of the label, and of the model’s

predictions to the grid resolution is examined in Section 3.10.

2.5.4 Spiking Neural Encoding

A spiking neural front-end based on the leaky integrate-and-fire (LIF) neuron [37] is included as an event-driven encoder; the neuron’s sub-threshold membrane potential evolves as

$$\tau_m \frac{dV}{dt} = -(V - V_{\text{rest}}) + R I(t), \quad (7)$$

where V is the membrane potential, τ_m the membrane time constant, V_{rest} the resting potential, R the membrane resistance, and $I(t)$ the input current. A spike is emitted whenever the potential crosses threshold,

$$S(t) = \begin{cases} 1, & V(t) \geq V_{\text{th}}, \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

after which the potential is reset. In discrete time, the dynamics are realised as

$$V_t = (\lambda V_{t-1} + I_t) (1 - S_t), \quad (9)$$

where $\lambda \in (0, 1)$ is the membrane leak (decay) factor and the factor $(1 - S_t)$ resets the potential to zero after a spike (hard reset, as implemented). The non-differentiable spike function is back-propagated through using a surrogate-gradient approximation [38]; spiking neural networks have recently shown promise for EEG decoding and emotion recognition [39–41]. In the full model the LIF stack does not receive the raw EEG time series. In parallel with the graph attention it receives a learned band-weighted sum of the per-window band-power node features of Section 2.5.2, one 19-electrode vector per 0.5 s window, and is run for ten time steps, one per window (two LIF layers of 128 units); its spike trains are layer-normalised before the next layer, and its embedding is concatenated with the graph embedding and projected to the EEG representation. A separate configuration, in which the same encoder is applied to the raw 1,500-sample epoch pooled to 300 temporal steps, each aggregating five consecutive 300 Hz samples (≈ 16.7 ms per step), is used only in the positive control of Supplementary Table S16, where the question is what

the spiking front-end alone extracts from EEG temporal dynamics. The encoder was included to provide an event-driven representation with an inductive bias toward salient transients over tonic background activity. Whether that bias helps is not assumed: its effect within the full model is measured by ablation (Section 3.4), its cost on conventional hardware by direct timing (Supplementary Table S9), and its capacity on its own by a positive control in which the encoder is trained alone on raw EEG (Supplementary Table S16).

2.5.5 Hierarchical Dynamic Graph Attention Learning

The dynamic functional graphs are processed by graph attention [42]. For connected nodes i and j with feature vectors h_i and h_j , an unnormalised attention coefficient is computed as

$$e_{ij} = \text{LeakyReLU}(a^\top [W h_i \parallel W h_j]), \quad (10)$$

where W is a shared learnable projection, a is a learnable attention vector, $\parallel$ denotes concatenation, and LeakyReLU is the leaky rectifier. Coefficients are normalised over each node’s neighbourhood with a softmax to yield α_{ij} , and node representations are updated by attention-weighted aggregation,

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W h_j \right), \quad (11)$$

where σ is a non-linear activation and $\mathcal{N}(i)$ is the neighbourhood of node i . Applied hierarchically over the dynamic graphs, this mechanism enables the network to learn which electrodes and which functional connections are most informative for engagement, adaptively and on a per-window basis. Graph attention, rather than a spectral or fixed-aggregation graph convolution, is used because the learned coefficients α_{ij} double as an interpretable importance signal (Section 3.11) without requiring a separate post-hoc attribution step, and because attention naturally accommodates the per-window adjacency changes of Eq. (4) without retraining fixed convolutional filters for each graph topology. The internal graph attention network (GAT) baseline (Table 4), which applies the same attention mechanism of Eqs. (10)–(11) to

a single static graph per epoch, performs at chance on this label (balanced accuracy 0.50, MCC 0.01), and so, within the margin of the paired tests, does the proposed gaze-free branch built around it (0.55; Table 4). Graph attention is therefore retained for the per-window adjacency changes it accommodates and for the descriptive importance maps it yields, not for a demonstrated accuracy gain.

2.5.6 Temporal Transformer Encoder

To model dependencies across time windows, the sequence of graph-level embeddings is passed through a transformer encoder [43] whose core operation is scaled dot-product attention,

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (12)$$

where Q , K , and V are the query, key, and value matrices and d_k is the key dimensionality, with $\sqrt{d_k}$ scaling the dot products to stabilise gradients. Multi-head self-attention allows the encoder to capture long-range temporal dependencies across the epoch without the sequential bottleneck of recurrent models, complementing the local, event-driven dynamics that the spiking encoder emphasises. A self-attention transformer, rather than a recurrent temporal encoder (long short-term memory, LSTM, or gated recurrent unit, GRU), is used because the sub-window sequence produced by the 5 s epoch (Section 2.4) is short, so the quadratic self-attention cost of Eq. (12) remains inexpensive (Section 2.5.10) while every pair of sub-windows can interact directly in a single layer, rather than through the sequentially-propagated hidden state of a recurrent model, which is more susceptible to attenuating the influence of early sub-windows on later ones.

2.5.7 Cross-Modal Fusion Strategy

Instead of concatenating modality features naïvely, the fusion module, termed the NeuroFusion Transformer in Fig. 2, integrates the EEG, graph, and ET representations in two stages. In the first stage, the three modality embeddings are treated as a short token sequence and processed by multi-head self-attention, through which the modalities exchange information mutually: the

EEG token attends to the eye-tracking token and the eye-tracking token attends to the EEG token, and both interact with the connectivity-graph token. In the second stage, a directed cross-attention refines the EEG representation by letting it query the connectivity-graph and eye-tracking context, keeping the fused embedding anchored on the EEG stream; this EEG-to-context direction is also what lets the gaze-free EEG branch be recovered by disabling the eye-tracking input entirely (Table 4). The EEG-to-context cross-attention is

$$Z_{\text{EEG} \rightarrow \text{ET}} = \text{Attention}(Q_{\text{EEG}}, K_{\text{ET}}, V_{\text{ET}}), \quad (13)$$

where the EEG representation provides the queries and the ET representation provides the keys and values, letting neural activity attend to behaviourally salient gaze information; an analogous EEG-to-graph cross-attention operates on the dynamic-graph embedding. The fused representation concatenates the EEG token with the two cross-modal embeddings,

$$Z_{\text{fusion}} = [Z_{\text{EEG}}; Z_{\text{EEG} \rightarrow \text{graph}}; Z_{\text{EEG} \rightarrow \text{ET}}], \quad (14)$$

retaining the EEG evidence while adding the learned conditional dependence between gaze, connectivity and cortical activity. A learnable gated residual fusion, $g = \sigma(W_g[\text{eeg} \parallel \text{cross_graph} \parallel \text{cross_et}])$, selectively passes modality information based on the current input, preventing fusion collapse when one modality is uninformative. The fusion and downstream reasoning pathway is illustrated in Fig. 4.

2.5.8 Neuro-Symbolic Reasoning Module

To introduce interpretable structure while keeping the model end-to-end differentiable, the neuro-symbolic module, termed the Neuro-Symbolic Decision Layer in Fig. 2, replaces opaque classification with an attention-weighted mixture of soft rules, in the spirit of interpretable-by-design and concept-based models [44, 45] and of neuro-symbolic learning [46]. A mixture-of-rules formulation, rather than a single global decision tree or a post-hoc surrogate explainer, is adopted

CROSS-MODAL FUSION AND NEURO-SYMBOLIC REASONING

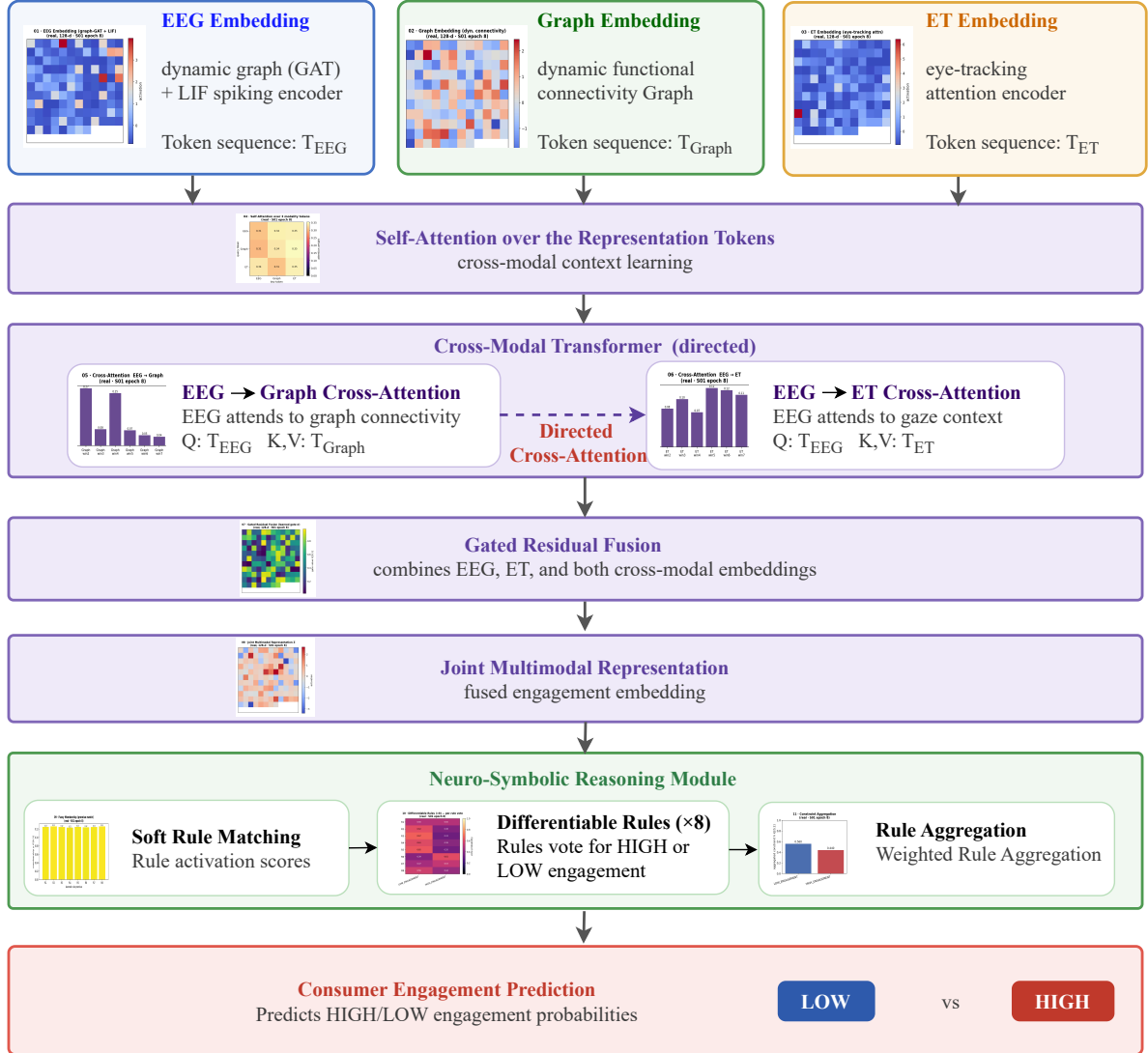


Fig. 4 Cross-modal fusion and neuro-symbolic reasoning: self-attention over the EEG, graph and eye-tracking tokens, directed cross-attention with the EEG representation as query, gated residual fusion, and the eight-rule neuro-symbolic module. In the production model the learned gate of Eq. (19) bypasses the rule pathway (Section 3.11).

because it remains differentiable end-to-end (permitting joint training with the representation-learning stack rather than a two-stage fit) while still yielding discrete, inspectable rule-activation traces at inference (Section 3.11); post-hoc surrogate explainers, by contrast, explain a separately-fitted approximation of the model rather than the model’s own computation [33]. The number of

rules, $n_r = 8$, is a fixed architectural hyperparameter, set a priori, rather than a value selected by a dedicated rule-count ablation; the diversity (L_{div} , Eq. (20)) and sparsity (L_{sparse} , Eq. (21)) regularisers instead constrain whatever number of rules is instantiated to remain distinguishable and sparsely activated, and formal sensitivity analysis of n_r itself is left to future work (Section 4.6). The

choice of eight is nonetheless a deliberate design decision rather than an arbitrary one, motivated by the following considerations:

- **Coverage of the engagement feature space.** Engagement here is driven by several partially independent signal families—frontal-theta and posterior-alpha power, functional-connectivity structure, gaze dispersion and dwell/ROI saliency, and pupil dynamics. Eight premises give enough distinct rules to associate with these complementary factors, whereas four would force several factors to share a premise and blur the resulting explanation.
- **Expressiveness versus auditability.** Eight rules are expressive enough to describe the HIGH/LOW decision structure yet few enough that the complete rule set is printed and inspected in a single panel (Section 3.11); a much larger bank would defeat the interpretability the module exists to provide, while too few would under-describe the decision.
- **Stable voting for a binary decision.** With each of the eight rules casting a HIGH/LOW vote, constraint aggregation was expected to form majority patterns without any single rule dominating. In the rule-only model this is what is observed: a minority of the rules dominate the great majority of epochs and the rest are rarely decisive (Section 3.11).
- **Robustness to the exact count.** The diversity (L_{div} , Eq. (20)) and sparsity (L_{sparse} , Eq. (21)) regularisers are intended to keep the instantiated rules distinguishable and sparsely activated, so that behaviour should not be sensitive to the precise value of n_r ; this expectation has not been tested by a rule-count sweep.
- **Interpretive coverage over fit.** n_r was chosen to maximise interpretive coverage rather than to fit the classifier. In the gated production model the module has no effect on accuracy because the gate removes it from the decision (Sections 3.4 and 3.11); in the variant trained with the gate closed, where the rules decide, eight rules cost two to three points of balanced accuracy relative to the gated model.

The fused representation $z \in \mathbb{R}^D$ is first projected into a rule-matching space,

$$k = \text{Dropout}(\text{GELU}(W_k z + b_k)), \quad (15)$$

where $W_k \in \mathbb{R}^{H \times D}$ and b_k are learnable and $k \in \mathbb{R}^H$ is the rule key. The module maintains n_r learnable rule premises $\{q_r\}_{r=1}^{n_r}$, $q_r \in \mathbb{R}^H$. Each rule fires in proportion to how well the input matches its premise, giving a soft activation

$$a_r = \frac{\exp(s_r/\tau)}{\sum_{r'=1}^{n_r} \exp(s_{r'}/\tau)}, \quad s_r = \frac{q_r^\top k}{\sqrt{D}}, \quad (16)$$

where s_r is the premise–key similarity, τ is a temperature (fixed at 1; not to be confused with the connectivity threshold of Eq. (4)), and $a_r \in [0, 1]$ with $\sum_r a_r = 1$. Each rule emits a class-level conclusion through its own linear head,

$$\ell_r = W_r k + b_r, \quad (17)$$

with $W_r \in \mathbb{R}^{C \times H}$ and C the number of classes. The rule conclusions are aggregated by their activations,

$$R = \sum_{r=1}^{n_r} a_r \ell_r, \quad (18)$$

so $R \in \mathbb{R}^C$ is the soft-rule evidence. A direct bypass classifier acts as a residual stabiliser, blended with the rule evidence through a learnable gate,

$$\hat{y} = \text{softmax}(\alpha(W_b z + b_b) + (1 - \alpha)R), \quad (19)$$

where $W_b z + b_b$ is the bypass logit, σ is the logistic function, $\alpha = \sigma(\alpha_0)$, and $\alpha \in (0, 1)$ is learned (initialised at $\sigma(0.30) = 0.57$). As $\alpha \rightarrow 0$ the decision becomes purely rule-driven, whereas as $\alpha \rightarrow 1$ it reverts to the unconstrained network. Because this gate could in principle let the classifier ignore the rule evidence, its learned value, the agreement between the rule-only decision $\arg \max R$ and the gated decision, the performance obtained when α is forced to zero at inference, and a variant trained throughout with $\alpha \equiv 0$ are all reported in Section 3.11.

Two auxiliary terms regularise rule quality. A diversity term penalises redundant premises,

$$L_{\text{div}} = \frac{1}{n_r(n_r - 1)} \sum_{i \neq j} (\hat{q}_i^\top \hat{q}_j)^2, \quad \hat{q}_i = \frac{q_i}{\|q_i\|_2}, \quad (20)$$

and a sparsity term minimises the entropy of the activations so that few rules dominate each

decision,

$$L_{\text{sparse}} = -\frac{1}{B} \sum_{b=1}^B \sum_{r=1}^{n_r} a_{b,r} \log a_{b,r}. \quad (21)$$

The combined rule regulariser is $L_{\text{rules}} = \lambda_{\text{div}} L_{\text{div}} + \lambda_{\text{sparse}} L_{\text{sparse}}$, where B is the batch size and $\lambda_{\text{div}}, \lambda_{\text{sparse}}$ are weights. In the gated production model the activations nonetheless remained uniform at inference ($a_r = 0.125$ for every rule; Supplementary Table S13): the gate had removed the rules from the decision, and the two regularisers, which enter the objective with a small effective weight ($\lambda_{\text{div}} = \lambda_{\text{sparse}} = 0.01$, further scaled by the rule-regularisation weight $\lambda_{\text{rules}} = 0.02$ of Eq. (26)), did not counteract the collapse. In the rule-only model, where the rules are used, distinct rules dominate distinct subsets of epochs (Section 3.11). At inference, each rule is grounded in the physical input space rather than in the latent key space: the activation a_r is attributed to the electrode-by-band node features of Section 2.5.2, to the three eye-tracking streams and to the ROI cells by integrated gradients [53] (zero baseline, 16 steps), averaged over held-out epochs; the largest-magnitude entries of the resulting 19×5 electrode-by-band map, with their sign, are reported as the premise of rule r and its head W_r as the conclusion (Section 3.11).

2.5.9 Subject-Invariance and Auxiliary Objectives

Subject-independent decoding is constrained less by class separability within a participant than by distribution shift across participants: an embedding that separates engagement well for the training population need not transfer to an unseen subject. Three auxiliary objectives are therefore combined with the classification loss to shape the learned representation, each addressing a distinct facet of this shift. The contrastive and MMD terms are removed individually in the ablation of Section 3.4; the domain-adversarial term is not ablated separately, and the gaze-free branch of Table 4 is trained without the contrastive term, which needs the gaze embedding, and without the MMD term (the adversarial term is retained); because the MMD term is an EEG-only subject-alignment objective that does not itself require

gaze, its removal is a confound of the gaze-free comparison, which Section 4.7 notes and the MMD ablation of Table 8 bounds.

To discourage subject-specific idiosyncrasies from dominating the fused representation, a maximum mean discrepancy (MMD) penalty aligns the embedding distributions of different subjects within each mini-batch. Let $\mu_s = \frac{1}{|Z_s|} \sum_{z \in Z_s} \phi(z)$ denote the kernel mean embedding of subject s , where ϕ is the feature map of a characteristic Gaussian kernel and Z_s the subject’s embedding set. The penalty is

$$L_{\text{mmd}} = \frac{1}{|B|} \sum_{(s,s') \in B} \|\mu_s - \mu_{s'}\|_{\mathcal{H}}^2, \quad (22)$$

where B indexes the subject pairs compared. As implemented, the penalty is evaluated on the EEG embedding between the epochs of one subject and the pooled epochs of the other subjects in the mini-batch, and only when both sides contain at least eight epochs; at batch size 32, with about twenty augmented epochs per training subject, that condition is met in few mini-batches, so the term acts sparsely, which is consistent with its negligible ablation effect (Section 3.4). Minimising L_{mmd} draws the per-subject feature distributions together and so promotes the subject-invariant representation that LOSOCV rewards. The MMD penalty is paired with a second alignment term of equal weight ($\lambda = 0.10$ each): a domain-adversarial subject classifier trained on the fused embedding through a gradient-reversal layer, in the manner of domain-adversarial neural networks; its loss L_{adv} is the cross-entropy of that classifier, and the reversal strength follows the standard schedule from 0 to 1 over training. The kernel-based moment-matching term was included because it optimises a single non-adversarial loss and is therefore less exposed to the min-max instability to which adversarial critics are prone when the number of samples per subject-domain is small, as within each LOSOCV mini-batch [30, 31]; the adversarial term was retained from development. Removing the MMD term costs 0.031 in balanced accuracy, not significant (Section 3.4).

Alignment alone does not tie the EEG representation to the concurrent behaviour. A cross-modal contrastive term (InfoNCE) therefore aligns the ℓ_2 -normalised EEG and eye-tracking

embeddings of the same epoch: for anchor i the positive is that epoch’s own eye-tracking embedding z_i^{et} , the negatives are the eye-tracking embeddings of the other epochs in the batch, weighted by their similarity to the anchor (hard-negative weights w_k), and with temperature $\tau_c = 0.07$,

$$L_{\text{con}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{e^{z_i^{\text{eg}\top} z_i^{\text{et}} / \tau_c}}{\sum_k w_k e^{z_i^{\text{eg}\top} z_k^{\text{et}} / \tau_c}}, \quad (23)$$

which involves no label: it makes the EEG embedding predictive of the concurrent gaze embedding, and is the training-time route by which gaze information shapes the neural pathway (Section 1).

An ROI-guidance term that keeps the eye-tracking attention distribution a consistent with the normalised region-of-interest saliency r read from the gaze stream,

$$L_{\text{roi}} = \text{KL}(r \parallel a), \quad (24)$$

was implemented as a further prior. On the training path used for every reported run, however, this term is evaluated only at inference and contributes no gradient, so it is not part of the effective objective; the “ROI” row of the ablation table (Section 3.4) accordingly refers to the ROI input pathway (the attention gate and the ROI graph modulation of Fig. 2), not to this term.

2.5.10 Model Optimisation and Training

The classification objective is the cross-entropy loss,

$$L_{\text{cls}} = -\sum y \log(\hat{y}), \quad (25)$$

where y is the ground-truth label and $\hat{y}$ the predicted probability; in practice a focal, class-balanced variant [47, 48] with label smoothing of 0.05 is adopted to counter class imbalance. Although the pooled 385-epoch dataset is close to balanced (193 High, 192 Low; Table 1), the per-fold training pool is not: each fold withholds one participant’s epochs entirely, so the realised HIGH/LOW ratio in training and, in particular, in the small held-out test set (approximately nine epochs per subject) varies across folds and is frequently skewed; the focal term down-weights already-well-classified majority-class epochs so that gradients are not dominated by whichever

class happens to be locally over-represented in a given fold. The complete training objective combines this classification loss with the neuro-symbolic rule regulariser of Section 2.5.8 and the auxiliary objectives of Section 2.5.9,

$$L_{\text{total}} = L_{\text{cls}} + \lambda_{\text{rules}} L_{\text{rules}} + \lambda_{\text{con}} L_{\text{con}} + \lambda_{\text{mmd}} L_{\text{mmd}} + \lambda_{\text{adv}} L_{\text{adv}}, \quad (26)$$

where the non-negative weights $\lambda_{\text{rules}}, \lambda_{\text{con}}, \lambda_{\text{mmd}}, \lambda_{\text{adv}}$ were fixed a priori after preliminary development and held constant across all folds; they were not tuned per fold or on the held-out subject, and the contrastive and rule terms are ramped in linearly over the first 40% of training. Because the dynamic functional graphs are precomputed per window (Section 2.5.2) and are not updated by gradients, no graph-regularisation term enters the effective training objective, and the ROI-guidance term of Section 2.5.9 is inactive on the training path. The contrastive and MMD terms are removed individually in the ablation of Section 3.4. Models were trained with the optimiser, learning rate, and schedule given in Table 3; early stopping monitored validation balanced accuracy rather than F1, so as not to reward class collapse. All hyper-parameters were held fixed across folds.

The overall computational cost is dominated by three operations. Dynamic functional-graph construction requires pairwise connectivity estimation across the $|V| = 19$ EEG electrodes, that is, $\mathcal{O}(|V|^2)$ per window; the hierarchical graph attention scales with the number of retained edges, $\mathcal{O}(|E|)$; and the temporal and cross-modal transformers incur the standard $\mathcal{O}(T^2)$ self-attention cost in the sequence length T , which stays small because each epoch yields only a short sequence of sub-windows. On the hardware used here the spiking encoder executes a dense matrix product at every leaky integrate-and-fire time step and is slower than a dense twin of the same widths (Section 3.4, Supplementary Table S9); an accumulate-only cost proportional to the firing rate would apply only on event-driven hardware, on which the encoder has not been run. All experiments were implemented in Python 3.12 using PyTorch 2.7.1 with CUDA 12.6, with each LOSOCV fold completing within practical training times. Each fold’s prediction is the probability

average of a 5-member deep ensemble (base random seed 42, applied to the Python, NumPy, and PyTorch generators; ensemble members differ by initialisation seed), which also provides seed-averaged stability. All hyperparameters, optimiser settings, and training schedules are listed in full in Table 3 to support independent reproduction.

To characterise deployment-time cost concretely, single-epoch inference latency was separately benchmarked for the full 1.72-million-parameter model (100 forward passes, after 10 warm-up passes, wall-clock timed) on a commodity CPU (AMD Ryzen 7, 8 cores/16 threads, 3.3 GHz; a separate PyTorch 2.11 installation, no GPU), distinct from the GPU environment used for training. At batch size 1, matching a real-time single-epoch inference scenario, latency was 14.9 ± 1.3 ms per epoch (67 epochs/s); batching improved throughput to 156 epochs/s at batch size 32. This CPU figure is arguably more informative for the on-device deployment scenario discussed in Section 4.5 than a data-centre GPU figure would be, since it does not rely on GPU acceleration being available at inference time.

2.6 Experimental Protocol and Evaluation Strategy

All models were evaluated under leave-one-subject-out cross-validation (LOSOVC): in each fold, every epoch from one held-out participant formed the test set, while epochs from all remaining participants formed the training pool. No held-out subject epochs were used as model inputs for parameter fitting or hyperparameter selection, yielding a subject-held-out estimate of subject-independent generalisation, the property required for prospective neuromarketing deployment; the dependence introduced by the pooled label construction (Section 2.4) is a separate question, quantified directly by the fold-wise-label robustness analysis of Section 3.10. Subject-independent evaluation was selected because random or subject-mixed data partitions allow recordings from the same participant to appear in both the training and test sets, potentially inflating performance through subject-specific characteristics rather than genuine task-related representations. Leave-one-subject-out cross-validation provides a more realistic estimate of generalisation

to previously unseen individuals and was therefore adopted as the primary evaluation protocol throughout this study. Performance is summarised as the mean $\pm$ standard deviation (SD) across folds for accuracy, balanced accuracy (BalAcc), precision, recall, F1-score and macro-averaged F1-score (Macro-F1), ROC-AUC, area under the precision-recall curve (PR-AUC), and Matthews correlation coefficient (MCC). Two operating points are reported for every model, and neither uses a test label. At the first, tabulated throughout as the uncalibrated operating point, the ensemble-averaged probability is thresholded at the Youden- J value found on a single held-out validation subject drawn from the training pool and never the test subject (the transferred threshold; values 0.05–0.73 across folds); because the validation participant is drawn without excluding the five single-class participants (Section 2.4), the threshold rule falls back to 0.5 whenever the validation set has fewer than four samples or only one label, which occurred in 2 of the 37 reported folds (test subjects S19 and S36, whose drawn validation participants, S33 and S41, are single-class). At the second, the calibrated operating point, the ensemble-averaged logits are divided by a temperature fitted on the same validation subject and thresholded at 0.5; because a validation subject supplies fewer than the twenty epochs the re-tuning step requires, the calibrated threshold is 0.5 in all but one fold, so this operating point coincides with a fixed 0.5 cut-off on the raw probabilities. The two differ only in the decision threshold; the ranking of epochs, and hence ROC-AUC, is identical. All eighteen internal baseline models (the eight EEG encoders of Table 4 and BrainGCN, three eye-tracking encoders and six fusion architectures; BrainGCN, at balanced accuracy 0.47 and ROC-AUC 0.47, is held out of the eight-encoder family of Table 4 for continuity with the submitted comparison and enters only the Holm family of Supplementary Table S10) were trained and evaluated on the same LOSOCV test folds, with one training-pool subject drawn at random per fold as the validation subject (an independent draw in the baseline runner, so not necessarily the subject used for the proposed model), the same early-stopping rule on validation balanced accuracy and the same two operating points. The protocols are matched in effort rather than identical. Each baseline was

Table 3 Implementation details: architecture, optimiser, and training hyperparameters held fixed across all folds.

Component	Configuration
Epoch length	5.0 s
Batch size	32
Learning rate	5×10^{-4}
Optimiser	AdamW
Training epochs	at most 250
Dropout	0.25 (0.17 in the eye-tracking encoder, 0.08 in the ROI, fusion and rule blocks)
Weight decay	0
Schedule	Cosine annealing with warm restarts ($T_0 = 50$ epochs, $\eta_{\min} = 10^{-6}$); gradient-norm clipping at 1.0
Early stopping	Validation balanced accuracy, patience 50, minimum 20 epochs
Transformer heads	4
Hidden / embedding dim	128
GAT heads (layer 1 / 2)	4 / 1
Connectivity threshold τ	0.30 (sensitivity: Section 3.10)
ROI saliency grid	5×2 (10 cells; sensitivity: Section 3.10)
Rule bypass gate α	learned (sigmoid); fidelity: Section 3.11
Loss function	Focal cross-entropy, $\gamma = 3$, class-balance factor $\alpha_{\text{focal}} = 1$ (distinct from the rule-bypass gate α), label smoothing 0.05, effective-number class weights [48]
Random seed (base)	42
Ensemble members	5 (seed-averaged)

tuned per fold: twelve hyperparameter configurations (learning rate, weight decay, dropout and batch size, drawn from an architecture-specific search space that always includes the previously used common budget of Adam, 10^{-3} , 10^{-4} , 32) were each trained as a single model with class-weighted cross-entropy and early stopping on the validation subject’s balanced accuracy (patience 30, at most 250 epochs), and the configuration with the highest validation balanced accuracy was the one applied to the test subject. The proposed model’s hyperparameters were fixed a priori and not searched per fold (Table 3), and it additionally uses a five-member ensemble, the focal loss and the training-time augmentation described below, which the baselines do not. Selection never touched the test subject, and the selected baseline hyperparameters are reported per fold in the repository. Results under the earlier common fixed budget (100 epochs, no early stopping, no search) are retained in Supplementary Table S8 for reference.

The cohort is small for models of this size (385 epochs; 37 evaluable participants; 1.72 M parameters), and overfitting was guarded against at several levels. Every fold trains on roughly 740 epochs after the deterministic half-epoch circular-shift augmentation, with stochastic amplitude

scaling, additive noise, channel dropout and temporal masking applied per sample; dropout (0.25, scaled down in the eye-tracking and fusion blocks), the MMD and adversarial subject-invariance terms, and the seed-averaged ensemble regularise the fit; early stopping is driven by a held-out subject rather than by training loss; and no hyperparameter was selected on a test subject. The empirical safeguards are the LOSOCV protocol itself, the label-permutation test (Section 3.6), and a subject-count learning curve in which LOSOCV is repeated on random subsets of 10 to 37 participants (Section 3.10, Supplementary Fig. S5); the slope of that curve at the full cohort size is the only available, and necessarily uncertain, indication of how the pipeline would behave on larger cohorts. Expected calibration error (ECE) is computed as a reliability-diagram error over ten equal-width probability bins (mean predicted probability versus empirical positive frequency per bin), pooled over all 347 held-out epochs into a single out-of-fold estimate, because a held-out subject contributes about nine epochs and a ten-bin ECE on nine observations is not meaningful; the Brier score, which needs no binning, is reported beside it. Both are computed from the saved held-out probabilities of every model with the same script

(`scripts/analysis/pooled_calibration.py`), so the calibration columns of Tables 4–6 are directly comparable; per-fold ECE values are not reported.

To facilitate reproducibility, the manuscript reports the complete preprocessing pipeline, model architecture, hyperparameter settings, evaluation protocol, statistical analyses, and ablation configurations. Subject-independent data partitioning was applied consistently across all experiments, and every reported performance metric was computed exclusively from held-out test subjects without exposure during model training.

3 Results

Having established the methodology and the leakage-aware evaluation protocol, this section examines whether INS-HDGS-CMT delivers reliable subject-independent engagement prediction. The analysis proceeds from overall predictive performance to component-wise validation, statistical significance testing, explainability, and detailed case studies.

3.1 Classification Performance

Under LOSOCV, INS-HDGS-CMT reached a mean accuracy of $78.24 \pm 16.60\%$, precision of $69.91 \pm 34.50\%$, recall of $77.18 \pm 32.37\%$, F1-score of $69.06 \pm 29.18\%$, and AUC of 0.88 ± 0.17 , with a balanced accuracy of $74.75 \pm 18.45\%$ and an MCC of 0.49 ± 0.36 (averaged over the 37 evaluable held-out subjects; the run of the production configuration under the 19-electrode montage of Section 2.1, whose fold checkpoints serve every analysis of this paper). These figures correspond to the model’s uncalibrated operating point, the threshold transferred from the validation subject (Section 2.6); at the calibrated operating point (temperature fitted on the validation subject, threshold 0.5) the counterparts were accuracy $76.66 \pm 16.25\%$, balanced accuracy $71.92 \pm 19.39\%$, and MCC 0.45 ± 0.38 , with AUC unchanged at 0.88 (threshold-invariant). The difference between the two is a threshold effect, the ranking being unchanged; the pooled expected calibration error of the held-out probabilities over the 347 epochs before temperature scaling (Section 2.6) was 0.14 and the pooled Brier score 0.18. Subject-level bootstrap 95% confidence interval (CI)s

were $[0.82, 0.93]$ for AUC, $[0.69, 0.81]$ for balanced accuracy, and $[0.37, 0.60]$ for MCC, and a within-fold label-permutation test on the saved held-out predictions (5000 shuffles of the labels within each held-out subject; `scripts/analysis/label_permutation_test.py`) gave $p = 2 \times 10^{-4}$ for mean per-fold AUC, balanced accuracy and MCC alike (null distributions centred on 0.50, 0.50 and 0.00, respectively). This test shows that the association between the held-out predictions and the labels is not attributable to chance; because the labels are shuffled against fixed predictions, it does not test whether the training procedure could fit random labels. The pooled receiver-operating-characteristic curve, the per-fold metric distribution, and the comparison against the label’s own EEG and gaze terms appear in Fig. 5. These figures describe subject-independent decoding of the rule-defined index of Section 2.4, not of engagement measured independently of it: the AUC reflects ranking of HIGH-index against LOW-index epochs, and Sections 3.2 and 3.5 locate its source in the gaze inputs.

3.2 Comparison with Internal Baseline Models

A deliberate evaluation design governs the comparisons that follow. The NeuMa engagement labels are defined from a fixed weighted composite of EEG band-power terms and gaze statistics (Section 2.4, Supplementary Table S5), so neither modality is independent of the label. To make this explicit, a subject-grouped linear probe on the ten defining features (logistic regression, leave-one-subject-out) recovers the label with mean per-fold ROC-AUC 0.67 (pooled 0.61) from the five EEG terms alone, 0.92 (pooled 0.94) from the five gaze terms alone and 1.00 from all ten (Supplementary Table S15); the per-fold means are the floor against which every learned model below, all of which are likewise reported as per-fold means, should be read. Two consequences follow. The label is almost entirely gaze-determined, so a model that sees only EEG must learn beyond the rule’s weak EEG component to exceed 0.67, and the EEG-encoder ranking is informative to the extent that it does so. Conversely, because five gaze summaries already recover the label at 0.92, no eye-tracking or multimodal score below that value, including the full model’s 0.88,

LOSOCV performance on the engagement index (37 evaluable subjects)

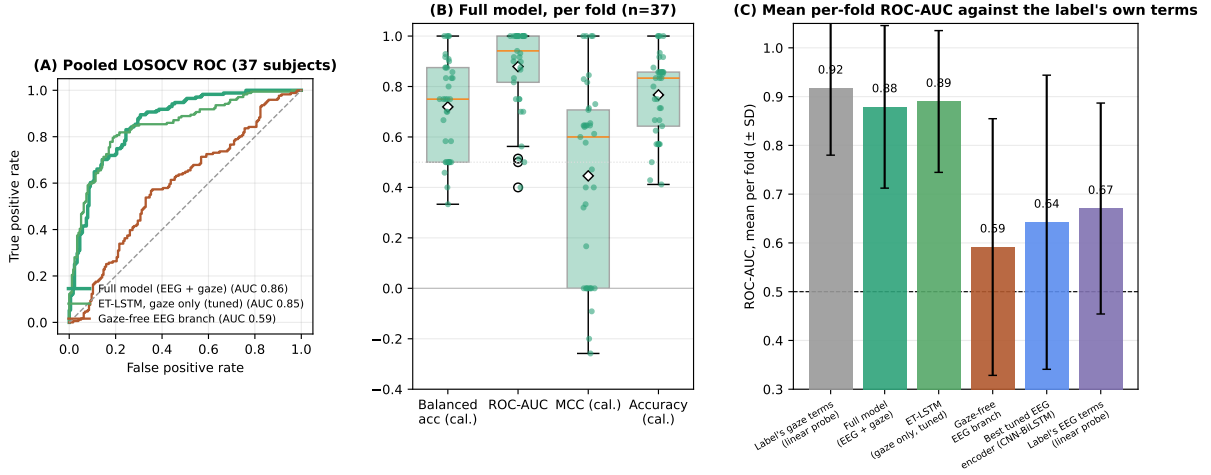


Fig. 5 LOSOCV performance (37 subjects). (A) Pooled ROC curves of the full model, the tuned gaze-only ET-LSTM and the gaze-free EEG branch. (B) Per-fold distributions of the full model’s metrics at the calibrated operating point (diamonds: means). (C) Mean per-fold ROC-AUC ($\pm$ SD) of the label’s gaze and EEG terms under a linear probe (Supplementary Table S15) beside the full model, the tuned ET-LSTM, the gaze-free branch and the best tuned EEG encoder.

can be read as information beyond the labelling rule; those scores are reported for completeness and as a modality-information analysis only. The EEG encoders, which receive no gaze input, supply the primary subject-independent ranking (Table 4), whereas the eye-tracking and multi-modal fusion comparisons (Tables 5–6) quantify the complementary information each modality contributes. All models were trained under identical LOSOCV folds, using the same dataset, preprocessing, and label-free post-hoc calibration procedure, and every baseline was tuned per fold by a nested validation-subject search under the same early-stopping rule as the proposed model, whose hyperparameters were fixed a priori (Section 2.6); Tables 4–6 report these tuned baselines, and Supplementary Table S8 the earlier common-budget values. Unless stated otherwise, metrics in Tables 4–6 are given at the uncalibrated operating point, with ECE the pooled out-of-fold reliability error and Brier the pooled Brier score (Section 2.6); the ablation analysis of Section 3.4 uses the same uncalibrated operating point.

Among EEG encoders (Table 4), the proposed gaze-free branch, dynamic graph attention with leaky integrate-and-fire spiking encoding and no gaze-derived input of any kind, reached balanced accuracy 0.55, ROC-AUC 0.59 and MCC 0.09

(37 folds). The eight tuned baselines spanned balanced accuracy 0.48–0.59 and ROC-AUC 0.49–0.64, with ShallowConvNet (0.59 / 0.62), DeepConvNet (0.59 / 0.63) and CNN-BiLSTM (0.57 / 0.64) numerically above the proposed branch. None of the eight paired comparisons was significant on any metric after Holm correction (balanced accuracy: $p_{\text{Holm}} \geq 0.78$; MCC: all 1.0; ROC-AUC: ≥ 0.44 ; Section 3.6, Supplementary Table S6), and the effect sizes were small (matched-pairs $|r_{\text{rb}}| \leq 0.38$; no comparison exceeded 24 wins against 12 losses over the 37 folds). The submitted version of this comparison reported the proposed branch as significantly better than every baseline. That result came from a branch that still received the gaze-derived ROI saliency vector through the ROI-attention gate (balanced accuracy 0.70, ROC-AUC 0.82, MCC 0.40 under the common budget) and from baselines trained under a fixed budget; with the ROI input removed and every baseline tuned per fold, the ranking collapses. On this label no EEG encoder, the proposed one included, reaches the 0.67 that the label’s own EEG terms recover linearly (the best, CNN-BiLSTM, 0.64); the earlier configuration is retained only as the “ROI-retained” variant of Section 3.5. That the EEG recordings do carry decodable signal was

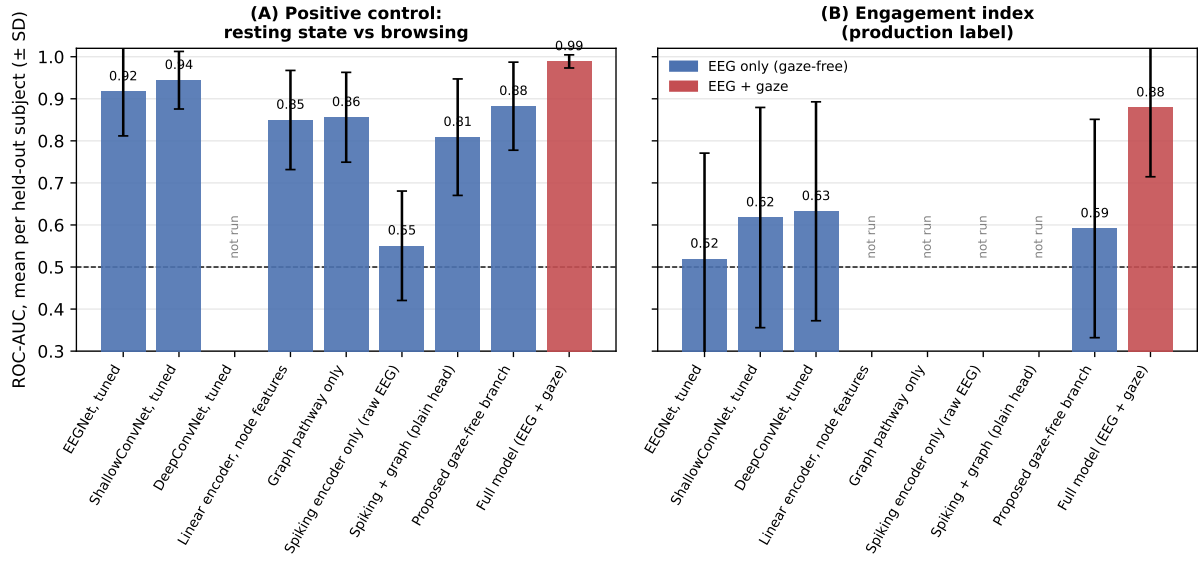


Fig. 6 The same EEG encoders on two labels from the same participants (Supplementary Table S16); bars are mean per-fold ROC-AUC $\pm$ SD. (A) Positive control, resting state versus browsing (42 folds). (B) The engagement index (37 folds); red marks the only encoder that sees gaze (the full model, 0.88). Blank positions are rungs not run on that track.

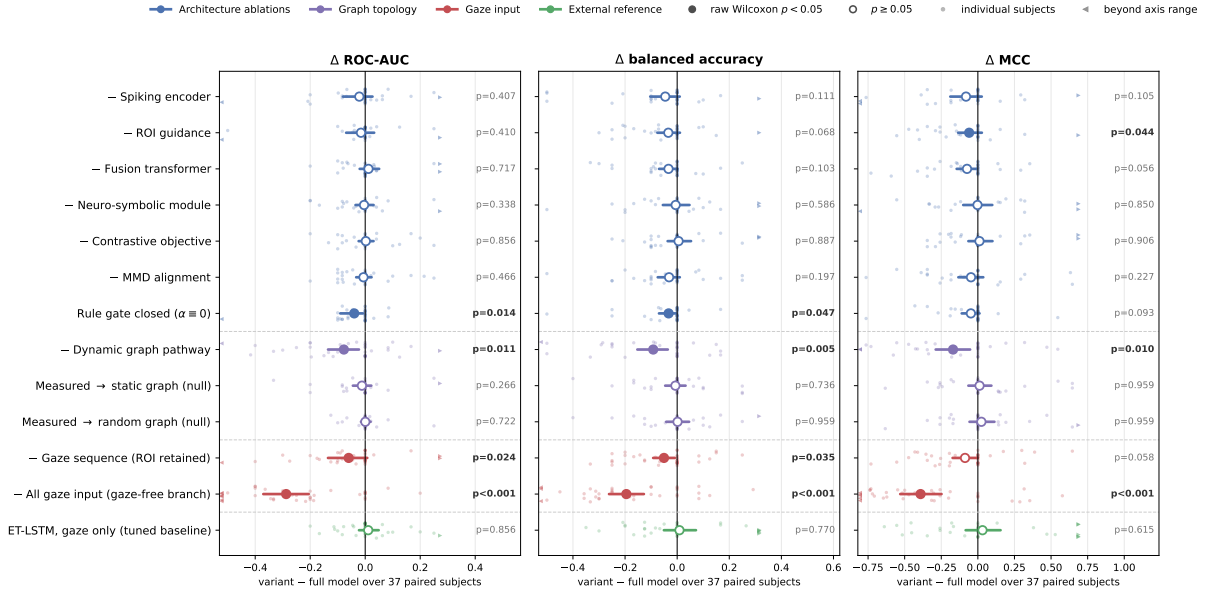


Fig. 7 Every control against the full model: mean paired difference over the 37 shared subjects with its 95% bootstrap CI (10,000 resamples); small points are individual subjects, triangles those beyond the axis range. The raw two-sided Wilcoxon p (zero differences discarded, as in Table 8) is printed per row and filled markers denote $p < 0.05$; Holm-corrected values are in Table 8 and Supplementary Table S14. Balanced accuracy and MCC are at the uncalibrated operating point.

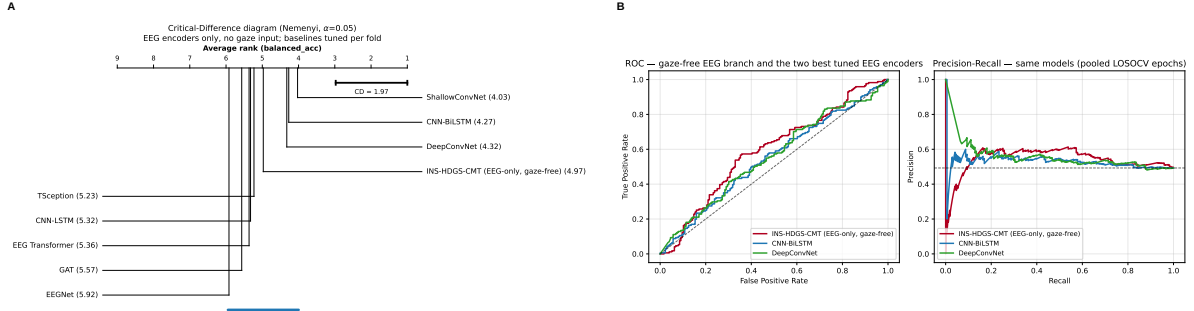


Fig. 8 Gaze-free EEG comparison, baselines tuned per fold. (A) Nemenyi critical-difference diagram over the nine EEG encoders (average rank on balanced accuracy, 37 folds; critical difference 1.97 at $\alpha = 0.05$). (B) Pooled ROC and precision-recall curves of the gaze-free branch (mean per-fold ROC-AUC 0.59), the convolutional neural network with bidirectional LSTM (CNN-BiLSTM, 0.64) and DeepConvNet (0.63).

confirmed with a positive control on the same subjects, resting state versus brochure browsing (Fig. 6A; Supplementary Table S16): tuned EEG-Net and ShallowConvNet reach ROC-AUC 0.92 and 0.94 on that contrast, a linear encoder on the model’s own node features 0.85, the graph pathway alone 0.86, the spiking encoder alone 0.55, the spiking encoder in front of the graph pathway 0.81, and the proposed gaze-free branch in its production configuration 0.88 (balanced accuracy 0.72); the full model with its gaze input reaches 0.99, since eye movement alone separates browsing from fixation. The EEG path can therefore decode a contrast that is present in the recordings, a few points below the best convolutional encoder; its 0.59 on the engagement label reflects that label (the gaze-free row of Fig. 7: -0.29 in ROC-AUC against the full model), and the dynamic-graph spiking EEG path is not the source of the framework’s accuracy on it (Fig. 8). Fig. 6 sets the same encoders side by side on the two labels: the EEG pathway decodes the positive control and sits at the level of the label’s own EEG terms on the engagement index; on the third label recorded from these participants, the behavioural purchase label, every configuration is near chance within the epoch (ROC-AUC 0.61 for the full model; Supplementary Table S16, Section 4.5).

The eye-tracking-only encoders (Table 5) and multimodal fusion architectures (Table 6) are considered next. Because five of the ten label-defining terms are gaze statistics, every model that sees gaze sits close to the 0.92 that those terms recover linearly: the tuned eye-tracking encoders

reach balanced accuracy 0.75–0.77 and ROC-AUC 0.88–0.89, the fusion baselines 0.73–0.77 and 0.83–0.88, and the full INS-HDGS-CMT 0.75 and 0.88. Under Holm-corrected paired Wilcoxon tests over the 37 folds (Supplementary Table S10) no difference between the full model and any of the nine eye-tracking or fusion baselines survived correction on balanced accuracy, MCC or ROC-AUC (smallest $p_{\text{Holm}} = 0.68$, Late Fusion on ROC-AUC; matched-pairs rank-biserial $|r_{\text{rb}}| \leq 0.39$, with 15 to 21 of the 37 folds tied on ROC-AUC). The multimodal comparison is therefore read as parity: the full model reaches the ceiling that the gaze component of the label sets, and so does an ET-LSTM. Calibration does not favour the full model either: over the pooled 347 held-out epochs its reliability ECE is 0.14 and its Brier score 0.18, against 0.06–0.15 and 0.15–0.19 for the eye-tracking and fusion baselines (Tables 5–6), so most gaze-based baselines are as well or better calibrated.

3.3 Contextual Positioning Against Prior NeuMa Work

The controlled quantitative claims rest on the internal baselines evaluated under identical folds and labels (Section 3.2); because no prior study reports subject-independent engagement decoding on NeuMa, a strict head-to-head is not possible. For context, Table 7 situates the proposed model relative to the only published decoding study on the same dataset, the graph-signal-processing hybrid of Kalaganis et al. [18], through

Table 4 EEG encoders under LOSOCV with no gaze input to any model (baselines tuned per fold, Section 2.6); the gaze-free branch is listed first. Mean $\pm$ SD over 37 folds at the uncalibrated operating point; ECE and Brier pooled over the 347 held-out epochs. Common-budget values: Supplementary Table S8.

Model	BalAcc	Macro-F1	MCC	ROC-AUC	PR-AUC	ECE	Brier
INS-HDGS-CMT (EEG-only, no gaze input)	0.55$\pm$0.17	0.51$\pm$0.29	0.09$\pm$0.33	0.59$\pm$0.26	0.66$\pm$0.26	0.05	0.25
ShallowConvNet [50]	0.59 $\pm$ 0.19	0.47 $\pm$ 0.30	0.16 $\pm$ 0.37	0.62 $\pm$ 0.27	0.67 $\pm$ 0.27	0.22	0.30
DeepConvNet [50]	0.59 $\pm$ 0.21	0.48 $\pm$ 0.27	0.15 $\pm$ 0.39	0.63 $\pm$ 0.26	0.70 $\pm$ 0.24	0.29	0.36
CNN-BiLSTM	0.57 $\pm$ 0.15	0.40 $\pm$ 0.32	0.13 $\pm$ 0.32	0.64 $\pm$ 0.30	0.70 $\pm$ 0.27	0.20	0.32
CNN-LSTM	0.52 $\pm$ 0.17	0.44 $\pm$ 0.23	0.04 $\pm$ 0.35	0.57 $\pm$ 0.29	0.66 $\pm$ 0.26	0.26	0.35
EEG Transformer	0.52 $\pm$ 0.18	0.44 $\pm$ 0.30	0.02 $\pm$ 0.37	0.59 $\pm$ 0.28	0.67 $\pm$ 0.24	0.24	0.32
TSception [51]	0.52 $\pm$ 0.17	0.44 $\pm$ 0.23	0.02 $\pm$ 0.31	0.56 $\pm$ 0.24	0.64 $\pm$ 0.24	0.30	0.34
GAT	0.50 $\pm$ 0.15	0.33 $\pm$ 0.29	0.01 $\pm$ 0.31	0.49 $\pm$ 0.25	0.61 $\pm$ 0.25	0.07	0.26
EEGNet [49]	0.48 $\pm$ 0.18	0.42 $\pm$ 0.22	-0.04 $\pm$ 0.34	0.52 $\pm$ 0.25	0.62 $\pm$ 0.22	0.19	0.30

Table 5 Eye-tracking encoders under LOSOCV (tuned per fold); label-coupled, since five label terms are gaze statistics. Mean $\pm$ SD over 37 folds at the uncalibrated operating point; ECE and Brier pooled over the 347 held-out epochs.

Model	BalAcc	Macro-F1	MCC	ROC-AUC	PR-AUC	ECE	Brier
ET-Transformer	0.77 $\pm$ 0.19	0.71 $\pm$ 0.29	0.54 $\pm$ 0.39	0.88 $\pm$ 0.17	0.91 $\pm$ 0.15	0.07	0.16
ET-LSTM	0.76 $\pm$ 0.19	0.70 $\pm$ 0.30	0.52 $\pm$ 0.37	0.89 $\pm$ 0.15	0.91 $\pm$ 0.14	0.10	0.16
ET-GRU	0.75 $\pm$ 0.17	0.68 $\pm$ 0.29	0.51 $\pm$ 0.33	0.88 $\pm$ 0.16	0.90 $\pm$ 0.15	0.08	0.16

Table 6 Multimodal fusion models under LOSOCV (baselines tuned per fold); label-coupled, since both modalities enter the label. Mean $\pm$ SD over 37 folds at the uncalibrated operating point; ECE and Brier pooled over the 347 held-out epochs. Paired tests against the full model: Supplementary Table S10 (none survives Holm correction).

Model	BalAcc	Macro-F1	MCC	ROC-AUC	PR-AUC	ECE	Brier
Cross-Attention	0.77 $\pm$ 0.17	0.71 $\pm$ 0.27	0.54 $\pm$ 0.34	0.87 $\pm$ 0.15	0.88 $\pm$ 0.17	0.08	0.15
Multimodal Transformer	0.77 $\pm$ 0.16	0.68 $\pm$ 0.28	0.53 $\pm$ 0.31	0.84 $\pm$ 0.18	0.87 $\pm$ 0.19	0.15	0.18
DynamicGAT + ET Transformer	0.77 $\pm$ 0.19	0.70 $\pm$ 0.29	0.54 $\pm$ 0.37	0.88 $\pm$ 0.16	0.89 $\pm$ 0.17	0.07	0.16
Dual Transformer	0.76 $\pm$ 0.19	0.71 $\pm$ 0.28	0.54 $\pm$ 0.37	0.86 $\pm$ 0.17	0.89 $\pm$ 0.19	0.08	0.16
INS-HDGS-CMT (full)	0.75$\pm$0.18	0.69$\pm$0.29	0.49$\pm$0.36	0.88$\pm$0.17	0.90$\pm$0.17	0.14	0.18
INS-HDGS-CMT (w/o neuro-symbolic)	0.74 $\pm$ 0.17	0.67 $\pm$ 0.30	0.48 $\pm$ 0.34	0.87 $\pm$ 0.18	0.90 $\pm$ 0.17	0.17	0.18
Early-fusion multilayer perceptron (MLP)	0.73 $\pm$ 0.19	0.69 $\pm$ 0.30	0.46 $\pm$ 0.38	0.85 $\pm$ 0.19	0.88 $\pm$ 0.17	0.06	0.17
Late Fusion (CNN-LSTM + ET-LSTM)	0.73 $\pm$ 0.20	0.64 $\pm$ 0.33	0.45 $\pm$ 0.39	0.83 $\pm$ 0.18	0.88 $\pm$ 0.14	0.12	0.19

Cohen’s κ , the agreement-beyond-chance metric used in that work for this class-imbalanced setting. The two studies are not directly commensurable, differing in several respects: the prediction target (Buy/NoBuy purchase intent, labelled from mouse clicks, versus the present HIGH/LOW engagement), the EEG preprocessing (an 8–30 Hz band versus the 1–45 Hz band used here), the epoch construction (gaze-defined, variable-length epochs versus fixed 5 s windows), and the evaluation protocol (within-subject leave-one-epoch-out over all 42 subjects versus the stricter subject-independent LOSOCV over the 37 globally-evaluable subjects here). The comparison is therefore presented as positioning on the shared NeuMa data, not as a like-for-like ranking. Within these caveats, the prior graph-hybrid attains $\kappa = 0.35$; the gaze-free EEG branch reaches only $\kappa = 0.08$ (0.10 calibrated), and it is the label-coupled

full multimodal model, $\kappa = 0.45$ (0.41 calibrated), that shows comparable-to-higher chance-corrected agreement under the harder subject-independent protocol.

Subject-independent ROC-AUC is in any case a quantity prior NeuMa work did not report. The full multimodal model’s ROC-AUC is 0.88 (Section 3.1), which is not the highest of the models evaluated: the individually tuned gaze-only ET-LSTM reaches 0.89, and no difference among the gaze-based models survives correction (Section 3.2). It is also a label-coupled measure, because both modalities enter the label definition and serve as model inputs (Section 4.1), whereas the gaze-free EEG branch, the only input configuration independent of the label’s gaze terms, reaches ROC-AUC 0.59 (Section 3.2). Cohen’s κ , F1, and balanced accuracy, by contrast, are bound

Table 7 Contextual Cohen’s κ against prior NeuMa work; targets and protocols differ, so this is positioning rather than a like-for-like comparison.

Study / method	Cohen’s κ
<i>Kalaganis et al. [18] — Buy/NoBuy, within-subject LOOCV</i>	
EEG indices	0.05
GFT-EEG	0.18
Hybrid indices	0.25
ET indices	0.27
GFT-hybrid (prior best)	0.35
<i>INS-HDGS-CMT (this work) — HIGH/LOW engagement, LOSOCV</i>	
Gaze-free EEG branch (no gaze input)	0.08 (0.10)
EEG branch, ROI vector retained (submitted configuration)	0.36 (0.43)
Full multimodal model	0.45 (0.41)

to a single operating point and to the per-subject threshold transfer of Section 3.8.

3.4 Ablation Analysis

To quantify each component’s contribution, one module at a time was removed and re-evaluated under identical LOSOCV folds (Table 8), recording the paired change in balanced accuracy and MCC alongside a Wilcoxon signed-rank test against the full model. Among the architectural ablations, removal of the graph pathway produced the largest degradation (balanced accuracy -0.092 , raw $p = 0.005$; MCC -0.169 , raw $p = 0.010$; after Holm correction over the eight original ablations the balanced-accuracy reduction remained significant, adjusted $p = 0.041$, the MCC reduction did not, $p = 0.077$). Whether this reflects the *measured* dynamic connectivity was then tested directly with two density-matched nulls in which node features, ROI vector and gaze were left untouched and only the adjacency was replaced: a static graph (the k most persistent edges of the epoch, k the mean edge count per window, repeated over all windows) and, per window, an Erdős-Rényi graph with the same number of edges onto which the epoch’s edge weights were re-dealt (the graph attention consumes only the binary adjacency of Eq. (4), so the correlation magnitudes are not used by it, and the connectivity term of the objective is evaluated on the precomputed weights and carries no gradient, Section 2.5.9; each null therefore changes the topology while preserving graph density). Neither null changed performance (balanced accuracy -0.007 , $p = 0.74$, and $+0.001$, $p = 0.96$; MCC

$+0.012$ and $+0.024$; Table 8). Fig. 7 places every control of this section and of Section 3.5 on a common axis with bootstrap intervals, so that the size of each effect can be read against the others rather than from its p -value alone. The graph pathway therefore contributes through the additional processing it applies to the node features, not through the measured time-varying topology, and the submitted description of the dynamic functional graph as the accuracy-critical component is withdrawn in that form. The one input whose removal matters is gaze: removing every gaze-derived input costs 0.195 in balanced accuracy ($p < 0.001$) and 0.391 in MCC, in line with the label’s construction (Section 3.5). Training with the rule gate closed ($\alpha \equiv 0$) costs 0.033 in balanced accuracy ($p = 0.047$; Section 3.11). Removing the eye-tracking sequence branch, the spiking encoder, the ROI gating, the temporal fusion transformer or the MMD domain-alignment term reduced balanced accuracy by 0.03 to 0.05, none of them significant after Holm correction (adjusted $p \geq 0.25$), and removing the neuro-symbolic module or the contrastive objective left it unchanged; these reductions are of the size of the run-to-run variation of the full model (balanced accuracy 0.72–0.75 and ROC-AUC 0.86–0.88 over three trainings of the same configuration with the same seed). These components are retained for the distinct capabilities they provide rather than for predictive gain, and the limits of their accuracy contribution are stated explicitly. The spiking encoder provides a biologically inspired temporal representation and an event-driven processing mechanism, but its direct contribution to

predictive performance is not demonstrable (balanced accuracy -0.046 , raw $p = 0.11$, adjusted $p = 0.52$), and no claim of predictive necessity is made for it. The positive control shows the cost more directly: trained on its own on the raw EEG epochs (300 leaky integrate-and-fire steps, 42 folds), the spiking encoder separates resting state from brochure browsing at ROC-AUC 0.55, a contrast that EEGNet and ShallowConvNet decode at 0.92 and 0.94 on the same recordings, so the spiking front-end discards most of the signal that convolutional encoders use; placed in front of the graph pathway with a plain head it lowers that pathway’s 0.86 to 0.81, a loss that the production configuration of the branch (0.88) makes up through its regularisers and ensemble rather than through the encoder. No energy advantage is claimed. The trained spiking encoder operates at a mean firing rate of 7.5% on the fold-1 checkpoint timed in Supplementary Table S9 and 10.6% on the fold-1 checkpoint of the submitted run (about 90% spike sparsity). Such sparsity would reduce energy only on event-driven neuromorphic hardware, on which the encoder has not been run and to which we had no access, so the hardware-energy claim of the submitted version is withdrawn rather than estimated; an operation-count projection would in any case ignore the layer normalisation applied to the spike trains between the LIF layers, which would itself have to be removed for an accumulate-only implementation. On the conventional hardware actually used, every LIF time step still executes a dense matrix product, and direct measurement shows that the spiking encoder is slower than a dense twin of identical layer widths in which each LIF layer is replaced by a linear-GELU block: 2.70 ms versus 0.45 ms per single-epoch inference on the commodity CPU of Section 2.5.10 (3.45 versus 0.69 ms at batch size 32), and 3.99 versus 0.41 ms (4.03 versus 0.43 ms at batch size 32) on an NVIDIA A100; on the A100 neither encoder raised the board’s power measurably above its 85 W idle draw, so the incremental inference energy could not be resolved and only latency is reported (Supplementary Table S9). The encoder is therefore retained as an event-driven representation that is compatible with, and would only pay off on, neuromorphic deployment; its efficiency rationale should be read as prospective rather than demonstrated.

The remaining ablated terms are auxiliary regularisers or structural components whose small, non-significant effects are consistent with the roles they were designed for. The cross-modal contrastive objective and the MMD domain-alignment term both act on the representation geometry rather than directly on the classification boundary: they encourage, respectively, agreement between the EEG and gaze embeddings of the same epoch and closer alignment of per-subject embedding distributions, refinements that the primary cross-entropy objective and the dynamic-graph representation already capture to first order, so their removal is not expected to move balanced accuracy sharply. Removing the eye-tracking *sequence* branch alone produces a modest, non-significant reduction because the ROI saliency vector that remains already carries most of the gaze information the label rewards; it is the removal of all gaze-derived input, the EEG-only row, that removes the accuracy (Section 3.5). Removing the ROI attention gate and graph modulation, while the ROI vector still reaches the eye-tracking branch, changes nothing; Section 3.5 shows that it is the ROI vector as an input, not the gating it drives, that carries accuracy. Removing the fusion transformer, the cross-modal attention mechanism that integrates EEG, graph, and eye-tracking representations (Section 2.5.7), also leaves balanced accuracy statistically unchanged; this is consistent with the parity of the fusion design with the other multimodal architectures (Section 3.2): that comparison tests the fusion mechanism against fundamentally different competing designs, whereas the present ablation removes it from a pipeline in which the gaze inputs already supply most of the discriminative structure. With the tuned baselines (Supplementary Table S10) the two analyses agree: the cross-modal transformer is neither better than the simpler fusion designs of Table 6 after correction nor an indispensable component of this architecture.

The neuro-symbolic module likewise primarily contributes interpretability and rule-based reasoning rather than improvements in predictive accuracy. Removing the module did not change balanced accuracy (Wilcoxon $p = 0.59$; -0.006 balanced accuracy, -0.002 MCC; Table 8), consistent with the module’s intended role as an explanation layer rather than an accuracy-enhancing one. The fidelity analysis of Section 3.11, however,

shows that in the gated model the rules do not take part in the decision, so their traces cannot be read as explanations of it. Explanations are therefore taken from the model trained with the gate closed ($\alpha \equiv 0$), in which the rules decide, at a cost of 0.033 in balanced accuracy at the uncalibrated operating point (Table 8) and 0.020 at the calibrated point (Section 3.11); interpretability is bought at that price rather than obtained for free. The one nuance is that the mean per-fold ROC-AUC in the multimodal comparison is marginally lower without the module (0.88 to 0.87; Table 6); this difference is small relative to the per-fold variability and does not affect the balanced-accuracy and MCC comparison of Table 8, but it is stated here rather than omitted.

3.5 Contribution of the Eye-Tracking Pathway

Because the framework is presented as cross-modal, the gain attributable to the eye-tracking pathway is isolated explicitly, by paired comparison over the same 37 held-out subjects of the full model with four reduced variants (Supplementary Table S14; uncalibrated operating point, Holm correction within each metric): the gaze *sequence* branch and cross-modal transformer removed but the ROI saliency vector retained (the configuration originally tabulated as the “EEG branch”); the ROI vector removed but the gaze sequence retained; the cross-modal transformer replaced by a bidirectional EEG–eye-tracking cross-attention block without the graph token; and all gaze-derived input removed (the strictly EEG-only branch). Removing the gaze sequence lowered ROC-AUC by 0.060 (95% CI $[-0.009, 0.138]$; $p_{\text{Holm}} = 0.12$; $r_{\text{rb}} = 0.56$; 17 wins, 16 ties, 4 losses over folds), balanced accuracy by 0.051 ($[0.009, 0.092]$; $p_{\text{Holm}} = 0.21$) and MCC by 0.088 ($p_{\text{Holm}} = 0.22$), none significant after correction. Removing the ROI vector alone lowered ROC-AUC by 0.015 ($p_{\text{Holm}} = 1.0$), and replacing the cross-modal transformer by the cross-attention block changed it by 0.012 in the variant’s favour ($p_{\text{Holm}} = 1.0$). Removing every gaze-derived input is a different order of effect: ROC-AUC fell by 0.287 (95% CI $[0.207, 0.374]$; $p_{\text{Holm}} < 0.001$; $r_{\text{rb}} = 0.96$; 29 wins, 6 ties, 2 losses), balanced accuracy by 0.195 ($[0.129, 0.261]$; $p_{\text{Holm}} < 0.001$) and MCC

by 0.391 ($p_{\text{Holm}} < 0.001$). Two conclusions follow; the per-subject picture behind both is given in Supplementary Fig. S8. First, the eye-tracking pathway is not a complement to an EEG core but the carrier of the accuracy: what remains without it (ROC-AUC 0.59) does not exceed the label’s own EEG terms (0.67). Second, the neural pathway adds nothing that a gaze-only encoder lacks: against the individually tuned ET-LSTM the full model differs by -0.011 in ROC-AUC ($p_{\text{Holm}} = 1.0$, $r_{\text{rb}} = -0.05$), -0.009 in balanced accuracy and -0.032 in MCC (both $p_{\text{Holm}} = 1.0$). The submitted claim that the full model exceeded ET-LSTM by 0.055 in ROC-AUC ($p_{\text{Holm}} = 0.046$) was made against the fixed-budget encoder (ROC-AUC 0.85 on the fold-matched probabilities of that test); the tuned encoder reaches 0.89, and the difference disappears.

3.6 Statistical Validation

Table 9 presents the gaze-free significance analysis: the proposed EEG branch against each tuned EEG baseline on balanced accuracy, using paired Wilcoxon signed-rank tests over the 37 folds (zero differences discarded, as in every paired test of this paper) with Holm–Bonferroni correction and the matched-pairs rank-biserial correlation r_{rb} as effect size. None of the eight comparisons was significant after correction on balanced accuracy, MCC or ROC-AUC (raw $p \geq 0.05$; $p_{\text{Holm}} \geq 0.78$ on balanced accuracy, 1.0 on MCC, ≥ 0.44 on ROC-AUC; Supplementary Table S6), and the median paired differences were within ± 0.14 . A Friedman omnibus test over the nine tuned encoders rejected equality on balanced accuracy ($p = 0.024$) and MCC ($p = 0.030$) but not on ROC-AUC ($p = 0.27$); the differences it detects lie among the baselines themselves (ShallowConvNet and DeepConvNet against EEGNet and GAT), not between the proposed branch and any of them. The within-fold label-permutation test of Section 3.1 (5000 shuffles of the held-out labels against the saved predictions) gives $p = 2 \times 10^{-4}$ for the full model’s mean per-fold AUC, balanced accuracy and MCC (null distributions centred on 0.50, 0.50 and 0.00), so the association between its held-out predictions and the labels is not attributable to chance; it is not, on its own, evidence that the training procedure cannot fit random labels. The critical-difference diagram

Table 8 Component ablation under LOSOCV (uncalibrated operating point; full model balanced accuracy 0.747, MCC 0.485), each row paired against the full model. Paired two-sided Wilcoxon tests (zero differences discarded), Holm-adjusted within the balanced-accuracy and MCC families of the eight original ablations; the four rows outside that family carry raw p . “– Eye-tracking sequence branch” retains the ROI saliency vector; “– All gaze-derived input” is the gaze-free branch of Table 4; the graph-null rows replace only the adjacency by a density-matched static or random graph. *Adjusted $p < 0.05$.

Configuration	Balanced accuracy			MCC		
	Δ	p_{raw}	p_{Holm}	Δ	p_{raw}	p_{Holm}
Full model	—	—	—	—	—	—
– Dynamic graph	−0.092	0.005	0.041*	−0.169	0.010	0.077
– Eye-tracking sequence branch (ROI retained)	−0.051	0.035	0.25	−0.088	0.058	0.33
– Spiking encoder	−0.046	0.111	0.52	−0.081	0.105	0.42
– ROI gate and graph modulation	−0.034	0.068	0.41	−0.059	0.044	0.31
– Fusion transformer	−0.033	0.103	0.52	−0.074	0.056	0.33
– MMD alignment	−0.031	0.197	0.59	−0.047	0.227	0.68
– Neuro-symbolic module	−0.006	0.586	1.00	−0.002	0.850	1.00
– Contrastive objective	+0.005	0.887	1.00	+0.011	0.906	1.00
– All gaze-derived input (EEG-only)	−0.195	< 0.001	—	−0.391	< 0.001	—
Rule gate closed ($\alpha \equiv 0$, rule-only decision)	−0.033	0.047	—	−0.048	0.093	—
Measured graph → density-matched static graph	−0.007	0.736	—	+0.012	0.959	—
Measured graph → density-matched random graph	+0.001	0.959	—	+0.024	0.959	—

(Fig. 8A; nine encoders, 37 folds, critical difference 1.97 ranks) places all nine encoders within one critical difference of each other (ShallowConvNet best at an average rank of 4.03, the proposed branch fourth at 4.97, EEGNet last at 5.92), and the pooled ROC/PR curves (Fig. 8B) show the proposed branch and the two encoders with the highest mean per-fold ROC-AUC all close to the chance diagonal, the branch marginally above them on pooled epochs and below them per fold. The submitted version of this section reported seven of eight comparisons as significant; the difference is entirely accounted for by the ROI input that the earlier branch received and by the fixed baseline budget (Section 3.2).

To reduce the likelihood of spurious findings, statistical conclusions are supported using paired non-parametric hypothesis testing, multiple-comparison correction, bootstrap confidence intervals, effect-size estimation, and permutation analysis wherever appropriate.

3.7 Single-Marker Concordance: Engagement Is Encoded Multivariately

To test whether engagement on NeuMa is carried by the canonical univariate EEG markers, rather than only by the multivariate graph representation, each marker was evaluated directly against the labels with a within-subject label-permutation test (20000 permutations over 347 epochs from

37 subjects; Fig. 9A). No marker reached significance: the within-subject HIGH–LOW contrast was negligible for frontal-theta power (Cohen’s $d = +0.09$, $p = 0.18$), posterior-alpha power ($d = -0.03$, $p = 0.40$), and fronto-posterior phase-locking in both the theta ($d = +0.04$, $p = 0.72$) and alpha ($d = -0.04$, $p = 0.66$) bands; every observed effect lies within its permutation null and within the negligible-effect range ($|d| < 0.2$). Each marker thus discriminates the index at essentially chance (single-feature ROC-AUC ≈ 0.5). The same epochs are separated at ROC-AUC 0.59 by the gaze-free EEG branch and 0.88 by the full model (Fig. 9B). Two readings must be kept apart. The contrast between the null univariate markers and the full model shows only that the index is carried by its gaze terms, which the univariate EEG markers do not see. The contrast between the null markers and the gaze-free branch shows that whatever index-related structure the EEG carries is distributed across electrodes and bands rather than concentrated in any single canonical marker; it does not show that graph-structured integration is required to recover it, since a linear encoder on the same band-power node features matches the graph pathway on the positive control (Supplementary Table S16) and density-matched topology nulls match the measured graph on this label (Section 3.4). The analysis is a more stringent, hypothesis-testing treatment of the dataset’s neurophysiology than the descriptive connectivity summaries of prior NeuMa work [18]; it motivated

Table 9 Gaze-free EEG branch versus each tuned EEG baseline on balanced accuracy (37 folds; paired Wilcoxon, zero differences discarded, Holm-corrected over the eight baselines; r_{rb} = matched-pairs rank-biserial correlation, positive favouring the branch). No comparison is significant.

Baseline	Median Δ	p	p_{Holm}	r_{rb}
EEGNet	+0.050	0.097	0.78	+0.33
GAT	+0.083	0.122	0.85	+0.30
EEG Transformer	+0.000	0.322	1.00	+0.20
CNN-LSTM	+0.000	0.402	1.00	+0.16
TSception	+0.042	0.462	1.00	+0.14
CNN-BiLSTM	+0.000	0.482	1.00	-0.15
DeepConvNet	-0.050	0.344	1.00	-0.19
ShallowConvNet	-0.062	0.378	1.00	-0.17

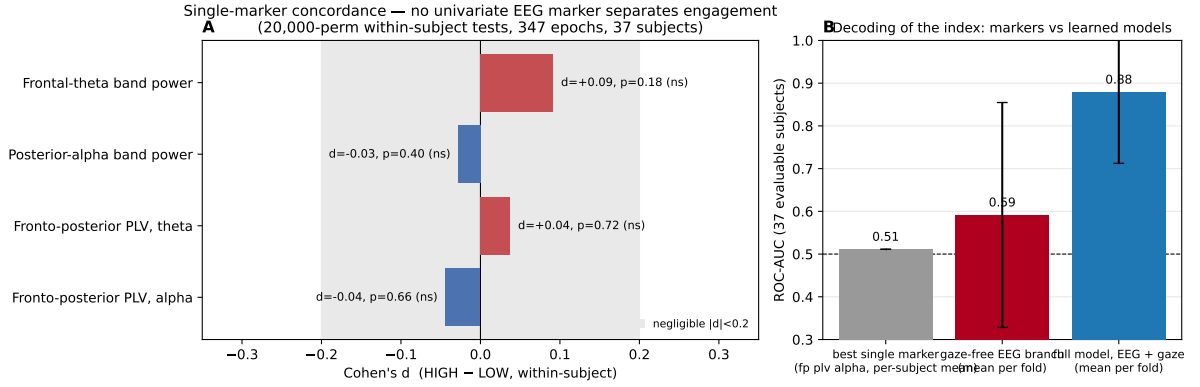


Fig. 9 No single marker carries engagement. (A) Within-subject effect size (Cohen’s d) of each canonical univariate marker with its permutation p (20,000 permutations); grey band $|d| < 0.2$. (B) ROC-AUC of the best single marker (fronto-posterior alpha PLV), the gaze-free branch and the full model on the same 37 subjects.

the dynamic-graph design, whose contribution the nulls then bound.

3.8 Subject-Wise LOSOCV Performance

Supplementary Fig. S1 reports fold-wise (subject-wise) performance. Most held-out subjects were decoded well above chance, but a minority of folds showed markedly lower accuracy, of a magnitude that the per-subject diagnostics of Section 3.9 attribute mostly to threshold transfer. These hard subjects, marked by per-subject probability-scale shifts and, in isolated cases, ranking inversion, account for most of the gap between mean and best-case performance and motivate the future directions of Section 4.6.

These probability-scale shifts are a calibration effect rather than a ranking failure: the model’s

ROC-AUC (0.88) comfortably exceeds what its transferred-threshold balanced accuracy (0.75) would imply. A label-free, prevalence-matched decision threshold per held-out subject, namely the quantile of the subject’s own *unlabelled* predicted probabilities at the training-set class prior, gave mean balanced accuracy 0.740 and MCC 0.43 (Supplementary Table S12), against 0.719 and 0.45 at the fixed 0.5 cut-off of the calibrated operating point and 0.747 and 0.49 at the transferred threshold; no test label enters this rule at any point. Because that rule uses all of a subject’s probabilities at once, it is transductive, and a deployed system would not have them in advance. A strictly causal variant was therefore evaluated in which the threshold for epoch k is computed only from the probabilities of the epochs that precede it, with the transferred validation-subject threshold used during a short unlabelled warm-up: with

a warm-up of three epochs (15 s of viewing) balanced accuracy was 0.748 and MCC 0.46, and with five epochs 0.746 and 0.47. After the warm-up, the label-free rule updates the threshold using only the new participant’s own preceding, unlabelled predictions; its initial value during the warm-up is inherited from the transferred validation-subject threshold, as stated above. The resulting operating point recovers that of the transferred threshold and sits 0.02 to 0.03 above the fixed 0.5 cut-off, so the adjustment is implementable prospectively, online, and without any label from the new participant. The per-subject z -score normalisation of the input (Section 2.2) is the one other step that uses statistics of the whole held-out recording; retraining the full model with per-epoch instance normalisation, which needs none, gave balanced accuracy 0.763 against 0.747 for the full model it is paired with at the operating point of Table 8 (Wilcoxon $p = 0.48$; MCC +0.03; 37 folds; 0.747 against 0.719, $p = 0.33$, at the calibrated point), so no part of the pipeline depends on seeing the new participant’s recording in advance. The change was heterogeneous across subjects (median $\Delta \approx 0$; Wilcoxon $p \geq 0.36$ for every variant against the fixed cut-off), so it is reported as a label-free threshold-adaptation analysis (a change of decision threshold, not of the predicted probabilities, and therefore not a probability calibration) rather than as the headline operating point; it is nonetheless consistent with, without establishing, a reading in which the residual balanced-accuracy gap reflects per-subject threshold transfer rather than a shortcoming of the learned representation. To avoid conflation, three balanced-accuracy operating points are reported for the same model, differing only in the decision threshold and never in the underlying ranking (ROC-AUC 0.88): 0.75 at the threshold transferred from the validation subject (the value tabulated throughout), 0.72 at the calibrated operating point of Section 2.6, which coincides with a fixed 0.5 cut-off, and 0.74–0.75 under the label-free per-subject thresholds analysed here; these are complementary threshold choices, not conflicting results.

3.9 Inter-Subject Variability and Failure Cases

To understand the lower tail of the fold distribution, a per-subject diagnostic was run on the

worst-performing held-out participants, decomposing each failure into its ranking, threshold, and calibration components. The analysis separates two qualitatively different modes. The larger group is threshold/calibration-limited but representationally sound: for a subset of subjects the model ranks the held-out epochs perfectly (ROC-AUC 1.00) yet scores only near-chance balanced accuracy at the transferred threshold; in each such case the diagnostic recommends a lower per-subject threshold and flags poor probability calibration. For these subjects the discriminative information is present and the error is one of operating-point transfer, consistent with the population-level calibration analysis of Section 3.8. A second, smaller mode reflects genuine representation failure, in which a held-out participant falls below chance ranking and is flagged as a subject-distribution shift, indicating that this participant’s EEG/eye-tracking characteristics are not well captured by the cross-subject model. That the majority of the hardest folds reflect threshold and calibration rather than discrimination, and only a minority reflect a true representation gap, supports the reading of Section 3.8 that the model’s limitation under a transferred threshold is largely a per-subject threshold problem. It also identifies the most promising remedy: label-free, subject-aware threshold adaptation (Section 3.8), with light subject-specific adaptation reserved for the small number of genuine distribution-shift cases; prospective subject-specific probability calibration remains future work.

3.10 Robustness to Label Definition, Connectivity Measure, Threshold, ROI Grid and Cohort Size

The global-threshold headline excludes five single-class subjects (S16, S31, S33, S41, S44) from the test folds, leaving 37 of 42 subjects evaluable (Section 2.4). To confirm that this exclusion does not bias the conclusions, and to obtain complete subject coverage, the EEG branch was re-evaluated under an identical model, preprocessing, channel montage, and LOSOCV protocol, altering only the label definition to a per-subject rank-based median. This split guarantees both classes for every participant and thereby renders

all 42 subjects evaluable. Because subject-median labels define engagement relative to each individual’s own distribution, they pose a within-subject discrimination task distinct from the absolute, cross-subject task of the global regime; the two are therefore regarded as complementary rather than interchangeable, with the global regime remaining the headline. Supplementary Table S1 lists both regimes side by side. The all-42 subject-median evaluation, run with the ROI-retained branch of the submitted version under the common budget (Supplementary Table S1), gives values comparable to the global regime for that configuration (balanced accuracy 0.69 over 42 subjects versus 0.70 over 37; ROC-AUC 0.80 versus 0.82), so excluding the five single-class subjects does not bias the comparison; the check has not been repeated with the gaze-free branch, whose global-regime values are 0.55 and 0.59 (Table 4). The modest reduction under the subject-median regime is expected, since it poses a harder, purely within-subject discrimination and additionally includes the five subjects that the global threshold leaves single-class. The check bears on subject coverage only; no conclusion about gaze-free EEG decoding rests on it, and Section 3.2 gives that result.

A second property of the label was tested by retraining rather than by re-scoring. Because the composite score is min–max normalised and thresholded over the pooled epochs of all participants (Section 2.4), each held-out participant contributes to the construction of its own labels. The full model and the gaze-free EEG branch were therefore retrained under a fold-wise label, in which the scaler and the median are fitted on each fold’s training participants only, together with a validation-participant rule that is identical for the proposed model and every baseline and never selects a single-class participant. All 37 folds and 347 held-out epochs remain evaluable under the fold-wise label, so no participant becomes single-class. Neither run changed materially (Supplementary Table S17): the full model gave balanced accuracy 0.715 against 0.747, ROC-AUC 0.872 against 0.879 and MCC 0.455 against 0.485, and the gaze-free branch 0.550, 0.605 and 0.091 against 0.552, 0.592 and 0.094. No paired difference approached significance (Wilcoxon $p \geq 0.32$) and every confidence interval of the mean difference includes zero. The largest change, 0.033 in

the full model’s balanced accuracy, is of the size of the run-to-run variation of the same configuration (0.718–0.747 over three trainings, Section 3.4), and the separation between the full model and the gaze-free branch remains of the same order (0.165 against 0.195 in balanced accuracy). The values reported throughout remain those of the pooled label, under which every control of Section 3.4 is paired; the fold-wise runs establish that the conclusions of Sections 3.2 and 3.6 do not rest on the pooled construction.

To assess sensitivity to the connectivity measure underlying the dynamic functional graph (Section 2.5.2), the EEG-only branch was re-evaluated under phase-locking value (broadband and five canonical-band variants: delta, theta, alpha, beta, gamma) in place of Pearson correlation, using a reduced subject subset ($N = 16$) and reduced ensemble/epoch budget for computational tractability. Under this reduced-scale protocol all seven connectivity measures showed near-chance discrimination (mean per-subject ROC-AUC 0.47–0.60, MCC clustered near zero; Supplementary Table S3), and at full 37-subject scale the gaze-free branch under Pearson correlation is itself near chance on this label (ROC-AUC 0.59, Table 4). The comparison therefore cannot favour any measure, and the topology nulls of Section 3.4 indicate why: on this label the model does not use the measured connectivity, whichever measure produces it. A full-scale comparison across measures would be informative only on a label for which the connectivity carries signal, and is left to future work (Section 4.6).

The connectivity threshold τ of Eq. (4) was examined at full scale (Supplementary Table S7, Fig. S4). Structurally, τ acts as a smooth density control: over the 3,850 windows of the cohort, the retained fraction of the 171 electrode pairs falls from 0.85 ($\tau = 0.10$) through 0.57 at the value used ($\tau = 0.30$; mean degree 10.3) to 0.15 ($\tau = 0.70$), with no discontinuity, whereas graph fragmentation, measured as electrodes left with fewer than three edges, stays below one electrode per window up to $\tau = 0.30$ (0.5) and then rises steeply (2.0 at $\tau = 0.40$, 4.9 at $\tau = 0.50$, 13.2 at $\tau = 0.70$); $\tau = 0.30$ thus sits at the last value before the graphs begin to fragment, which is the structural rationale for the a priori choice. Predictively, retraining the EEG-only branch under the full 37-subject LOSOCV

protocol for $\tau \in \{0.20, 0.25, 0.30, 0.35, 0.40, 0.50\}$ gave balanced accuracy 0.54–0.59 and ROC-AUC 0.52–0.59 under the reduced budget of the sweep (three-member ensembles, 120 epochs; at $\tau = 0.30$ this budget gives ROC-AUC 0.52 against the 0.59 of Table 4 under the production budget; Supplementary Table S7): the metrics are flat across the whole range, with no degradation at 0.50, and, consistent with Section 3.2, the gaze-free branch is near chance at every τ , so the threshold is structurally motivated and predictively immaterial.

The spatial resolution of the ROI representation was examined at two levels (Supplementary Table S11, Fig. S6). For the ROI saliency vector that the model consumes, rebuilding it on grids from 2×1 to 10×8 cells (including the 6×4 product layout of Fig. 1) changes its granularity as expected (mean largest-cell share 0.74 to 0.23) but preserves the per-epoch attentional-concentration ordering: the Spearman correlation of the largest-cell share with that of the 5×2 vector is 0.71–0.82 for every grid of six or more cells, and drops to 0.56 only for the two-cell grid. The engagement label itself (Section 2.4) contains no spatial grid, only band powers of the P3/C3/F3/Fz mean and scalar gaze-position statistics, so grid resolution cannot affect it. For the model’s predictions, the full model was retrained with the ROI vector rebuilt on 2×1 , 3×2 , 6×4 and 8×6 grids: balanced accuracy 0.72–0.76, ROC-AUC 0.88–0.90, none differing significantly from the 5×2 run (smallest $p = 0.53$, ROC-AUC on the 6×4 grid), so the ROI resolution does not affect the result.

Finally, to address the size of the cohort, LOSOCV was repeated on random subsets of 10, 16, 24, 30 and 37 participants (three draws each; Supplementary Fig. S5). Held-out ROC-AUC of the full model rose from 0.86 at 10 participants to 0.87 at 16 and 0.88 at 37 (balanced accuracy 0.68 to 0.74; three draws per size, three-member ensembles, 150 epochs), with a slope between 30 and 37 participants of 0.001 ROC-AUC and 0.002 balanced accuracy per additional participant. Within the observed range of 10 to 37 participants, ranking quality has plateaued by 16 participants and only the operating-point metrics still rise slowly. A subsampling curve cannot establish what a cohort larger than the one recorded would give, so no claim is made that performance is unlimited by cohort size; the curve shows only that, within this range, the gain per additional participant is small,

which is consistent with a ceiling set by the label’s gaze construction (Section 3.5) rather than by the amount of training data. Together with the label-permutation test and the subject-held-out early stopping, this bounds, without removing, the risk that the reported performance is an artefact of fitting a small cohort; extrapolation beyond 37 participants remains uncertain.

3.11 Explainability Analysis

A defining aim of INS-HDGS-CMT is interpretability, increasingly seen as essential for trustworthy neural decoding [52]. The model’s internal evidence is therefore inspected at several complementary levels, summarised in Figs. 10–11 and Supplementary Table S2. Throughout, attributions are read as associations learned by the model rather than as causal claims about neural mechanism.

3.11.1 EEG Graph-Attention and Importance Analysis

EEG electrode and connection importance was estimated from the dynamic graph-attention coefficients (Fig. 10A,B) and from integrated gradients (Section 3.11.4). The per-electrode attention received (Fig. 10A) and the full electrode-by-electrode attention matrix (Fig. 10B) concentrate on frontal and parietal electrodes and their functional connections, in line with the established roles of frontal theta and parietal alpha dynamics in attention and engagement [1, 8], and consistent with the integrated-gradients attribution that ranks posterior-alpha and frontal-theta power as the leading EEG inputs after the ROI saliency vector (Supplementary Table S2). Because density-matched static and random adjacencies reproduce the full model’s accuracy (Section 3.4), these coefficients describe where the trained model attends, not what its decision depends on.

3.11.2 Eye-Tracking Attention Analysis

The gaze evidence that drives high- versus low-engagement predictions was characterised behaviourally and at the population level by analysing the raw gaze and pupil streams across all 385 epochs (Fig. 11). High-engagement epochs are set apart by markedly broader, more exploratory

INS-HDGS-CMT explainability (S01, fold-01; A,B gated model; C,D rule-only model)

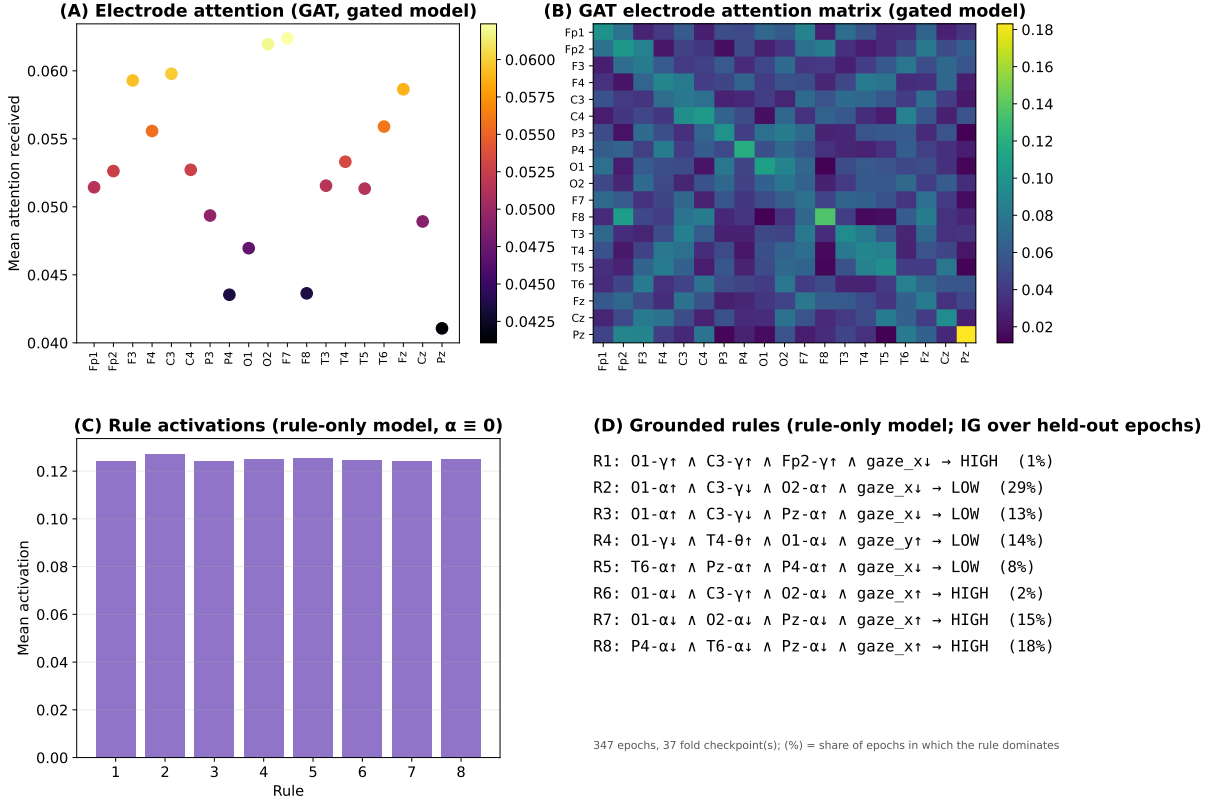


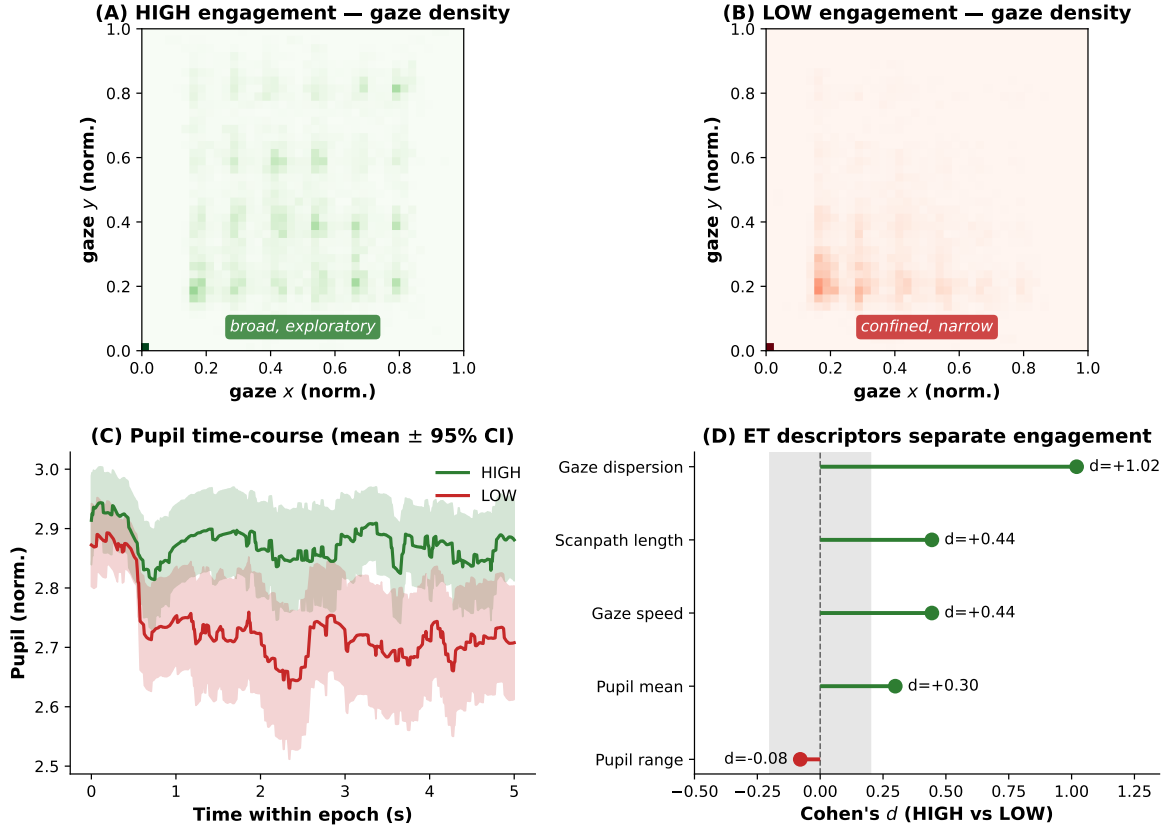
Fig. 10 Explainability for held-out subject S01. (A) Mean graph attention received per electrode and (B) the electrode-by-electrode attention matrix of the production model. (C) Mean soft-rule activations and (D) attribution-grounded rule premises (three largest electrode–band terms, dominant eye-tracking stream, rule conclusion) of the model trained with the rule gate closed (Section 3.11); full maps in Supplementary Fig. S7.

gaze, with greater spatial dispersion (Cohen’s $d = 1.02$), longer scan-paths, and faster gaze kinematics (both $d = 0.44$), together with modestly larger pupil dilation ($d = 0.30$) and sustained dwell on salient regions documented in the case studies (Section 3.12); low-engagement epochs show spatially confined gaze concentrated in a narrow region. Because the engagement label is itself derived in part from gaze statistics (Supplementary Table S5), these descriptors capture the behavioural signature of the label rather than independently predicting it, which is why the gaze-free branch of Table 4 is the only configuration whose score is independent of them.

3.11.3 Neuro-Symbolic Decision Explanations

The neuro-symbolic module produces per-epoch rule-activation traces (Fig. 10C) and, for each rule, a grounded IF–THEN summary (Fig. 10D). Each rule fires in proportion to how well the fused representation matches its premise and casts a vote for HIGH or LOW engagement, and the branch combines these votes in proportion to their activations following Eq. (18). Because the premises are vectors in a learned key space, they are not reported as latent coordinates; instead each rule’s activation is projected onto the physical inputs by integrated gradients (Section 2.5.8), yielding a 19×5 electrode-by-band attribution map per rule, of which the three largest-magnitude terms, with

Eye-tracking behavioural phenotype of engagement (NeuMa, 385 epochs: 193 HIGH / 192 LOW)



Descriptive phenotype: the engagement label is itself ET-derived, so these gaze/pupil descriptors characterise— rather than independently predict— engagement; the EEG comparison is the gaze-free evaluation.

Fig. 11 Eye-tracking phenotype (385 epochs). (A, B) Gaze-position density for HIGH and LOW epochs. (C) Pupil-diameter time course (mean $\pm$ 95% CI). (D) Cohen's d of the eye-tracking descriptors. The label is built partly from gaze statistics, so these descriptors characterise rather than predict engagement.

their sign, form an attribution-grounded summary of the rule, referred to below as its grounded premise (for example, rule 7: “O1- $\alpha \downarrow \wedge$ O2- $\alpha \downarrow \wedge$ Pz- $\alpha \downarrow$, with gaze- $x \uparrow \rightarrow$ HIGH”; full maps for all eight rules in Supplementary Fig. S7). The groundings are taken from the model trained with the gate closed (Section 3.11), whose rules decide; across its eight rules, 47–85% (mean 65%) of the attribution mass falls on the EEG node features and the remainder on the gaze and ROI inputs, every rule carries a horizontal-gaze term, and the electrode-band terms that recur are posterior alpha at O1, O2 and Pz, with C3 gamma: the three LOW rules that dominate 56% of epochs are alpha-up and gaze-left, the three HIGH rules that dominate 35% are alpha-down and gaze-right,

in line with the graph-attention and integrated-gradient importance of Sections 3.11.1 and 3.11.4 and with the label’s own gaze construction. These grounded premises summarise what the trained rules respond to; they are not the learned premises themselves, which remain vectors in the key space in which the matching is performed, and they are learned associations rather than neurophysiological mechanisms.

Whether the rule pathway is faithful to the decision, given the bypass gate of Eq. (19), was tested directly on the trained fold checkpoints (Supplementary Table S13). The learned gate never left its initialisation: $\alpha = 0.570$ (fold range 0.567–0.573; initial value 0.57), and the rule attention stayed uniform ($a_r = 0.125$ for all eight

rules, HIGH and LOW alike). The rule-only decision $\arg \max R$ agreed with the gated decision on 39% of held-out epochs (fold mean), and the rule margin was *negatively* correlated with the final logit margin ($r = -0.69$). Forcing α to zero at inference, so that the decision is taken from R alone without retraining, gave balanced accuracy 0.39 and ROC-AUC 0.22 against 0.72 and 0.88 for the gated decision, whereas forcing α to one (bypass only) gave 0.725 and 0.877, indistinguishable from the gated decision: in the gated model the rule pathway is not consulted, and its rule evidence is anti-aligned with the decision. A model trained throughout with the gate closed ($\alpha \equiv 0$) reached balanced accuracy 0.715 and ROC-AUC 0.839 against 0.75 and 0.88 for the full model at the uncalibrated operating point (Wilcoxon $p = 0.047$ and 0.014), which is the rule-gate row of Table 8 (-0.033 , $p = 0.047$); at the calibrated operating point the difference is 0.020 ($p = 0.15$), so the rules can carry the decision at a cost of two to three points. The conclusion is therefore split. The traces of the gated model are not explanations of its decision and are no longer reported as such; the traces and grounded premises of the rule-only model are, and Fig. 10D is drawn from that model, in which posterior alpha suppression at O1/O2/Pz together with a rightward gaze term characterises the HIGH rules and the reverse pattern the LOW rules. Quantitative assessment of rule stability across folds and of concept correspondence remains future work.

3.11.4 Attribution via Integrated Gradients

Feature attributions were computed with integrated gradients (IG) [53], which attribute a prediction to input feature i by integrating model sensitivity along a straight-line path from a baseline x' to the input x ,

$$\text{IG}_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha, \quad (27)$$

where F is the model output and α parameterises the interpolation path. Integrated gradients satisfy sensitivity and implementation-invariance axioms, providing axiom-consistent per-feature importance scores that were aggregated across folds to rank the most influential EEG and ET

features (Supplementary Table S2). The resulting ranking is dominated by the ROI saliency vector (0.52), with posterior alpha (0.18) and frontal theta (0.16) the leading EEG contributors and gaze position (0.13) the leading raw eye-tracking feature.

3.12 Case Studies of High and Low Engagement

To make the end-to-end decision process concrete, two representative held-out epochs are traced through the same trained ensemble: one HIGH-engagement exemplar and one LOW-engagement exemplar. Each is the most-confident, correctly-classified epoch of its class for that subject. The gaze trajectory, ROI saliency vector, decision and attribution appear in Fig. 12, the regional EEG signatures in Supplementary Fig. S2 and the two ROI saliency vectors side by side in Supplementary Fig. S3; the two workflows are compared in detail in the text below.

3.12.1 High-Engagement Workflow

For the HIGH exemplar, frontal-theta power is higher in the HIGH epochs than in the LOW epochs within the same subject, in keeping with a frontal attentional-control signature. The ROI saliency vector is spread over eight of the ten grid cells, both rows of the page included, and the gaze ranges over a wide spatial extent, signalling exploration of many products; the wide-ranging fixation pattern is consistent with the population-level attribution ranking (Supplementary Table S2), in which posterior alpha power, a marker of visual attention allocation via alpha suppression, is the second-most influential EEG feature after ROI saliency itself. Integrated-gradients attribution is distributed across modalities: the HIGH decision emerges from temporally coherent agreement between covert (EEG) and overt (gaze/ROI) attention.

3.12.2 Low-Engagement Workflow

For the LOW exemplar's subject, the within-subject frontal-theta contrast has the same sign (higher in that subject's HIGH epochs), whereas posterior alpha runs in opposite directions in the two subjects, echoing the population-level finding that neither marker separates the classes on

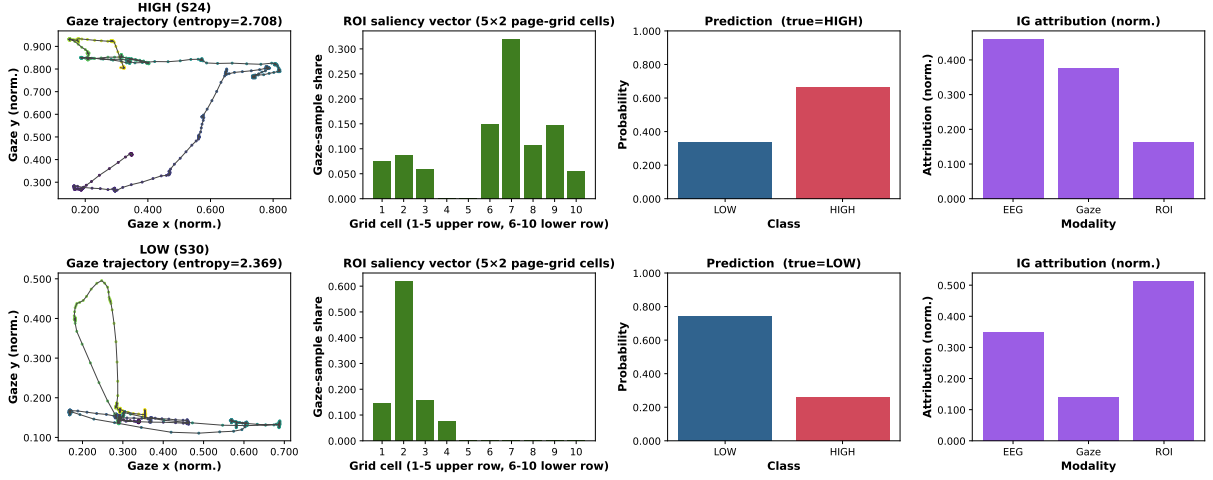


Fig. 12 HIGH (S24) and LOW (S30) exemplars: gaze trajectory, ROI saliency vector over the ten 5×2 grid cells (1–5 upper row, 6–10 lower row), predicted class probability, and integrated-gradients attribution over the three input streams that receive gradient (the connectivity input enters only as a binary mask and receives none).

its own. The defining feature is spatial: the ROI saliency vector is concentrated in four cells of the upper row (62% in a single cell) and is zero elsewhere, and the gaze stays confined to a narrow band of the page. Attribution is accordingly led by the ROI trajectory (about half of the attributed importance, with the EEG band-power features second): the LOW decision is driven chiefly by the confinement of fixation to one region of the page, mirroring the population-level attribution ranking (Supplementary Table S2), in which the ROI-saliency vector alone accounts for roughly half of total attributed importance (0.52), more than the next five features combined. Taken together, the two cases portray HIGH engagement as broad, multimodally coherent exploration of the page and LOW engagement as gaze confined to a single region, in keeping with the gaze-dispersion term of the label (Section 2.4) and the phenotype of Fig. 11, with frontal-theta a case-level EEG correlate that is null at the population level (Section 3.7) and ROI saliency the dominant single feature at both the population and case-study level.

4 Discussion

The controls locate the source of the reported accuracy; understanding what that implies

requires examining the learned representations, the multimodal interactions, and the individual architectural components. The following discussion interprets these findings in relation to the existing neuromarketing literature and the underlying neuroscience of consumer engagement.

4.1 Threshold-Free Discrimination as the Reporting Metric

Across the metrics examined, ROC-AUC is used as the headline metric for the decoding results, and the reasons are set out here before the accuracy is traced to its source in Section 4.2. Because both modalities enter the label definition, the gaze-free EEG-only branch is the only configuration whose score is independent of the label’s gaze terms; it reaches ROC-AUC 0.59 under LOSOCV, below what those EEG terms recover linearly (0.67) and indistinguishable from tuned EEG encoders (Section 3.2), so no EEG-only result is offered as a primary finding. The full multimodal model reached an ROC-AUC of 0.88 (95% CI [0.82, 0.93]; permutation $p = 2 \times 10^{-4}$), but because both modalities enter the label definition and serve as model inputs, this figure is interpreted as a secondary, label-coupled measure rather than an independent prediction

of engagement. AUC is the appropriate headline for this problem for three reasons. First, it is threshold-invariant: it measures how consistently the model ranks an engaged epoch above a non-engaged one, independently of any single operating point, which matters here because the residual balanced-accuracy gap is a per-subject thresholding effect rather than a ranking failure (Section 3.8). Second, ranking quality is what a neuromarketing application most often needs, ordering stimuli, segments, or audiences by predicted engagement, rather than a hard HIGH/LOW call at a fixed cut-off. Third, because LOSOCV makes the task subject-independent and the per-subject class balance is heterogeneous, with some held-out participants strongly skewed toward one engagement state, threshold-bound scores are volatile across folds, whereas AUC, by integrating over all operating points, remains comparatively stable under this imbalance. For these reasons, threshold-independent discrimination under subject-independent LOSOCV is the metric on which the decoding results are compared; the accuracy, F1, balanced-accuracy, and MCC figures are reported alongside as operating-point summaries. ROC-AUC is a reporting choice, not evidence for the architecture: what the discrimination rests on is the subject of the next section.

4.2 Where Does the Accuracy Come From?

The controls of Sections 3.2–3.5 allow the question to be answered component by component, and the answer is not the one the architecture was designed around. The accuracy comes from the gaze inputs. Removing every gaze-derived input costs 0.29 in ROC-AUC and 0.20 in balanced accuracy, whereas removing the gaze *sequence* while keeping the ROI saliency vector costs 0.06 in ROC-AUC and 0.05 in balanced accuracy, neither surviving correction, and removing the ROI vector while keeping the sequence costs nothing significant; either gaze input alone therefore carries most of what the label rewards, which is what its construction from five gaze statistics predicts. The full model reaches the same ceiling as a tuned eye-tracking LSTM (ROC-AUC 0.88 against 0.89), and no eye-tracking or fusion baseline differs from it after correction; the cross-modal transformer is

not what sets the level, since replacing it by a bidirectional cross-attention block leaves ROC-AUC 0.01 higher, not lower.

The EEG pathway contributes within that ceiling, not beyond it. Removing the graph pathway costs 0.09 in balanced accuracy, but replacing the measured dynamic adjacency by a density-matched static or random graph costs nothing at all (Table 8), so what the pathway adds is the additional processing it applies to the per-window node features, and the time-varying connectivity it was built to capture is not used. The gaze-free branch built on this pathway decodes the label at ROC-AUC 0.59, indistinguishable from eight tuned EEG encoders and below what the label’s own EEG terms reach linearly (0.67); on a contrast the EEG does carry, resting state versus browsing, a plain convolutional encoder decodes at ROC-AUC 0.92–0.94, a linear encoder on the same node features at 0.85, the spiking encoder alone at 0.55, and the gaze-free branch as a whole at 0.88. The spiking front-end therefore loses signal rather than adding a representation, and its removal from the full model is invisible only because the gaze inputs dominate. The neuro-symbolic module’s learned gate never leaves its initial value and its rule evidence is anti-aligned with the decision, so the rule traces of the gated model describe nothing about how it decides; a model trained with the gate closed does decide by its rules, at a cost of two to three points, and that model is the one from which explanations are drawn.

Read together, a simpler model, an eye-tracking encoder with or without a plain EEG branch, would reach the reported accuracy on this label. What the framework offers is not accuracy but the apparatus for finding this out: the recoverability audit, the gaze-free branch, the topology nulls, the fidelity test, the positive control and the behavioural label. These are reported in full, including the claims of the submitted version that they overturn, because a multimodal architecture evaluated without them would have been credited with mechanisms it does not use.

4.3 Positioning of the Engagement-Prediction Task

The binary HIGH/LOW engagement target differs from the Buy/NoBuy or like/dislike decisions

common in prior consumer neuroscience, so absolute scores are not strictly commensurable across tasks; the controlled claims accordingly rest on the eighteen internal baselines evaluated on identical NeuMa folds and labels, with the cross-study Cohen’s κ comparison (Section 3.3) serving only as contextual positioning. Within the gaze-free EEG comparison no encoder differs from any other after correction ($|r_{rb}| \leq 0.38$; Table 9), so no claim of practical significance is made for the EEG branch; the practically relevant result is the parity of the full model with tuned eye-tracking encoders (Supplementary Table S10), which means that a gaze-only model suffices for this index. Qualitatively, the findings agree with reports that EEG–ET integration aids engagement estimation [8] only in the sense that the index is itself constructed largely from gaze statistics, and they do not reproduce the benefit of connectivity features for consumer-EEG decoding reported elsewhere [12]: here, density-matched topology nulls perform identically to the measured connectivity.

4.4 Neuroscientific Interpretation of the Findings

The explainability analyses recovered structure that is plausible but has to be read against the label. The integrated-gradients attributions place the ROI saliency vector first and posterior alpha second (Supplementary Table S2), the grounded rules of the rule-only model combine posterior alpha at O1, O2 and Pz with a horizontal-gaze term in every rule (Section 3.11), and gaze dispersion and scan-path extent dominate the eye-tracking phenotype (Section 3.11), all in accordance with the attention literature [1, 8] and, more directly, with the five gaze statistics and five band-power terms of the P3/C3/F3/Fz mean from which the label is built. These remain learned associations, not demonstrations of causal mechanism. Read together with the single-marker concordance analysis (Section 3.7), in which each canonical univariate marker is individually null, the attributions indicate that the index is encoded multivariately across the band-power node features; the topology nulls of Section 3.4 and the linear encoder of Supplementary Table S16 show that this multivariate pattern is carried by the node features themselves, not by the connectivity between them.

At the level of individual cases, this multivariate pattern has a coherent neurophysiological narrative. In the HIGH-engagement exemplar, elevated within-subject frontal-theta power co-occurs with sustained, wide-ranging fixation and distributed EEG–gaze attribution (Section 3.12), consistent with an account in which frontal midline theta indexes top-down attentional control that is behaviourally expressed as continued, exploratory sampling of the stimulus [1, 8]. In the LOW-engagement exemplar, a weaker within-subject frontal-theta contrast co-occurs with an early collapse of gaze toward a narrow region, consistent with reduced attentional engagement rather than active but unsuccessful processing. This case-level narrative should not, however, be over-generalised: at the population level no single band-power or connectivity marker, including fronto-posterior theta/alpha phase-locking, discriminates engagement (Section 3.7). The most defensible reading is therefore that frontal-theta dynamics contribute to discrimination of the index as one element of a distributed pattern over electrodes and bands rather than as an independent, reliably synchronised marker in their own right. That reading motivated representing EEG as a dynamic graph; the topology nulls of Section 3.4 qualify it, since the distributed pattern is available from the node features alone and the measured connectivity adds nothing to it on this label.

4.5 Practical Implications for Consumer Analytics

The practical reading of these results is narrower than the framework’s design suggested. What the model estimates subject-independently is the NeuMa engagement index, a fixed rule over band powers of the P3/C3/F3/Fz mean and gaze statistics; whether that index tracks anything a practitioner would call engagement is not established here, and on the one behavioural criterion available, product purchase, neither the index’s inputs within a 5-s epoch nor the model predict choice (Supplementary Table S16). Within that limit, the model can be applied to new participants without per-person calibration, its ranking is threshold-invariant, and a light label-free per-user step can be run online from three to five unlabelled epochs (Section 3.8). Because a tuned eye-tracking encoder reaches the same accuracy

(Section 3.2), a deployment that has gaze available does not need the EEG pathway for this index, and where gaze is unavailable the EEG-only accuracy on the index (ROC-AUC 0.59) is not adequate for use. The inspectable traces that support transparent reporting to stakeholders and ethics boards [32] are those of the rule-only model (Section 3.11), not of the gated model. The CPU-only single-epoch latency of 14.9 ms (Section 2.5.10) permits real-time use; any energy advantage of the spiking front-end would require neuromorphic hardware and is not realised on the CPU or GPU used here (Section 3.4).

Beyond advertising evaluation itself, the same combination of properties, subject independence, calibrated ranking, and inspectable explanation traces, is what several adjacent applications require, so the framework is plausibly extensible, with appropriate re-validation in each target setting, to adaptive advertising systems that adjust content in response to estimated engagement, to engagement-aware human-computer interfaces more broadly, and to brain-computer-interface research that likewise seeks interpretable, subject-independent decoding from EEG. None of these extensions is evaluated here; they are noted as plausible directions consistent with the framework’s design goals, not as demonstrated capabilities.

4.6 Scope and Future Directions

This section closes by setting out the study’s scope and the directions it opens. The engagement labels follow a fixed rule-based definition, a weighted combination of five EEG band-power terms and five gaze statistics (Supplementary Table S5), so the study evaluates subject-independent decoding of that index rather than of an external behavioural criterion: the label’s construction is disclosed, its linear recoverability from each modality is quantified (Supplementary Table S15), the gaze-free EEG branch is the only configuration independent of the label’s gaze terms, and it decodes the index no better than tuned EEG encoders or the label’s own EEG terms. Anchoring engagement in a behavioural target was attempted with the purchase label of Supplementary Table S16, where the physiology within a 5-s epoch is uninformative and dwell time is not; a protocol that separates looking from

choosing, or self-report collected per stimulus, is the next step. The framework is validated on the NeuMa dataset alone, so cross-dataset validation on independent EEG–ET corpora is the principal route toward broad generalisability. Subject-independent decoding of the index under LOSOCV reaches the gaze ceiling on average, but a minority of hard-to-decode participants drives the per-fold variability; reliability- and subject-aware adaptation offers a concrete way to lift these cases. The reduced-scale connectivity-measure comparison of Section 3.10 was inconclusive at $N = 16$ subjects; a full-scale, 37-subject comparison between Pearson correlation and phase-locking value (broadband and band-limited variants) is identified as a specific direction for future work. Finally, the dynamic-graph, spiking, transformer, and symbolic stack is expressive, and lighter-weight encoders would extend it toward real-time deployment. Future work will therefore pursue behaviourally grounded labels, cross-dataset validation, reliability- and subject-aware adaptation, deployment-efficient encoders, richer symbolic knowledge bases, full-scale connectivity-measure comparison, and extension to graded or continuous engagement targets. A further direction opened by recent pretrained EEG transformers [54, 55], convolutional attention encoders [56] and adaptive tokenisation front-ends that embed EEG as discrete, inspectable tokens [57] is to substitute them for the dynamic-graph EEG branch within the same leakage-aware protocol, which would test whether representations learned on thousands of hours of unrelated EEG transfer to a 42-participant neuromarketing cohort under subject-independent evaluation. The learning curve of Section 3.10, which plateaus from sixteen participants, and the positive control of Supplementary Table S16, on which plain convolutional encoders already decode at ROC-AUC above 0.9, are, within the cohort sizes that could be tested, consistent with a ceiling imposed by the label construction rather than by the encoder: an index dominated by its gaze terms was not decoded from EEG beyond its EEG terms by any encoder tried here. Whether a larger cohort would change this remains untested.

4.7 Limitations

Although external validation would further strengthen the evidence for generalisability, comparable public datasets for subject-independent consumer engagement prediction using synchronised EEG and eye tracking are currently unavailable. To the best of our knowledge, NeuMa is the only publicly available multimodal neuromarketing dataset that combines synchronised EEG, eye-tracking recordings, and consumer engagement annotations acquired during a realistic consumer decision-making task. Evaluation was therefore performed under subject-held-out leave-one-subject-out cross-validation to provide the strongest subject-independent assessment possible within the available benchmark, and future work will validate the proposed framework on additional datasets as comparable multimodal neuromarketing resources become available.

Several further limitations qualify these conclusions. First, because the engagement label is a weighted composite of EEG band power and gaze statistics (Supplementary Table S5), no model is independent of it: the EEG encoders can in principle recover its EEG terms from their band-power node features, and the eye-tracking and fusion models its gaze terms, which is why the linear-probe recoverability of Supplementary Table S15 is reported as the floor against which learned performance is judged, and why the ranking among EEG encoders (Table 4) is anchored on models that receive no gaze input. Second, LOSOCV gives a small per-fold test set (about nine epochs per subject), so single-fold and hard-subject metrics are indicative rather than precise, and the reduced-scale robustness checks of Section 3.10 are correspondingly limited. Third, although every baseline was tuned per fold with a nested twelve-configuration search under the same validation-subject early-stopping rule as the proposed model, whose own hyperparameters were fixed a priori rather than searched (Section 2.6), the search spaces are architecture-specific and finite, so the absence of differences among the EEG encoders is a matched-effort result; a larger search could separate them, in either direction. Fourth, the cohort is small for models of this size; the learning curve of Section 3.10 and the subject-held-out early stopping bound the resulting overfitting risk but do not remove it, and generalisation to larger

cohorts is extrapolated from that curve rather than demonstrated. Fifth, the full model is more complex than any single baseline, and the results show that a simpler configuration, an eye-tracking encoder alone, suffices for this index. Sixth, the controls added in this revision, the topology nulls, the positive control and the behavioural label, were each run once under one configuration; they are sufficient to withdraw the claims they test but do not exhaust the design space of those claims, and the pooled-epoch label construction contributes a further shift of the same size when the held-out subject is excluded from it (Section 2.4); the run-to-run variation of the full model (about 0.03 in balanced accuracy and 0.02 in ROC-AUC over three trainings of the same configuration, Section 3.4) is of the same order as several of the ablation effects of Table 8. Seventh, the per-window band-power node features are estimated from 0.5 s windows (Welch segments of 128 samples at 300 Hz, 2.3 Hz bins), so the delta and theta bands rest on one to two spectral bins, a coarse resolution for the slow bands. Eighth, the engagement label is min-max scaled and thresholded at the median over the pooled epochs of all participants, so each held-out participant contributes to the construction of its own labels; refitting the scaler and median on the training participants only changes 9 of the 385 labels and lowers the re-scored ROC-AUC of the full model from 0.879 to 0.871 (Section 2.4); retraining the full model and the gaze-free branch under fold-wise labels changes neither run beyond the pipeline’s run-to-run variation (Section 3.10, Supplementary Table S17). The reported runs nonetheless use the pooled label, because the controls of Section 3.4 are paired within that protocol, so the fold-wise runs bound the effect of the pooled construction rather than remove it. Ninth, the validation participant used for early stopping and hyperparameter selection was drawn independently for the proposed model and for the baselines, so the two need not coincide within an outer fold, single-class participants were not excluded from that draw: for the reported production run this drew a single-class validation participant in 2 of the 37 folds (Section 2.6), for which the Youden-J threshold rule falls back to its 0.5 default rather than an ill-defined optimum, so the reported per-fold thresholds are unaffected in kind; early stopping’s validation balanced accuracy for those two folds

is nonetheless a less discriminating signal, since it cannot penalise a false call on the class absent from the validation participant. The proposed model is a five-member ensemble trained with augmentation and a focal loss whereas each baseline is a single model; the comparisons are matched in protocol but not in every nuisance factor, and no single-member run of the proposed model is reported. Tenth, the gaze-free branch of Table 4 was trained without the MMD term as well as without gaze, the adversarial subject term being retained (Section 2.5.9), so its comparison with the full model removes a subject-alignment regulariser together with the gaze inputs; the MMD ablation of Table 8 (-0.03 balanced accuracy, not significant) bounds this confound without removing it. As a consumer-neuroscience study, no clinical validity is claimed.

The dual-use character of engagement decoding from neurophysiological signals also warrants explicit acknowledgement. A model capable of estimating covert engagement from EEG and gaze could, in principle, be applied not only to evaluate advertising material but to optimise it for persuasive effect on individuals who have not consented to such profiling, or to infer engagement in contexts, such as political messaging, outside the commercial evaluation setting for which the NeuMa data were collected [34]. The present study did not, and does not, evaluate any such application; the model is trained and assessed exclusively on anonymised, consent-collected data (Section 2.1) for retrospective group-level analysis, and no individual-level, real-time, or covert deployment is proposed or endorsed. Responsible use of this class of methodology in applied neuromarketing settings requires informed participant consent, transparent disclosure of neurophysiological monitoring, and independent ethical oversight of any deployment beyond the retrospective research setting evaluated here, consistent with prior calls for ethical governance of consumer neuroscience [32].

Prospective validation against a behavioural or self-report criterion collected under a protocol that separates looking from choosing, on an independent cohort, remains the step that would turn subject-independent decoding of this index into evidence about engagement.

5 Conclusions

Predicting consumer engagement objectively and subject-independently from physiological signals remains a central challenge in neuromarketing, where self-report is biased and existing multimodal models are frequently leaky, naïvely fused, and opaque. We approached it with INS-HDGS-CMT, representing EEG as a sequence of dynamic functional graphs encoded with graph-attention and spiking modules and fused with eye tracking through a cross-modal transformer, and we subjected the framework to the controls that such architectures are rarely given. The principal findings are these. Under subject-held-out LOSOCV the full model decodes the NeuMa engagement index at ROC-AUC 0.88 and balanced accuracy 0.75, which is the ceiling set by the gaze statistics that define the label and is also reached by a tuned eye-tracking LSTM. The gaze-free EEG branch is statistically indistinguishable from eight tuned EEG encoders on this label, although the same EEG separates resting state from browsing at ROC-AUC above 0.9 with a plain convolutional network and at 0.88 with the branch itself. Removing the graph pathway lowers balanced accuracy by 0.09 (significant after correction), but the measured dynamic connectivity it is built on is not what it uses: density-matched static and random graphs perform identically. The learned rule gate bypasses the rules, and faithful rule explanations are available only from a model trained with the gate closed, at a cost of two to three points of balanced accuracy. On behavioural purchase labels, nothing inside a 5-s epoch of EEG or gaze predicts choice, whereas how long a product is looked at does. The spiking encoder is retained as an event-driven representation without predictive or, on the hardware used, efficiency benefit.

These results narrow rather than expand the claims of the submitted version, and we consider that the more useful contribution: the framework is a reproducible, leakage-aware reference for subject-independent multimodal engagement analysis, and the control battery, label recoverability, gaze-free branch, graph nulls, rule fidelity, positive control and behavioural labels, is the part of it that we recommend to future work. Behaviourally grounded labels collected under a protocol that separates looking from choosing, cross-dataset replication and pretrained EEG

encoders are the next steps toward engagement prediction that is more than the decoding of its own definition.

List of abbreviations. AI: artificial intelligence; BalAcc: balanced accuracy; CI: confidence interval; CNN-BiLSTM: convolutional neural network with bidirectional long short-term memory; CPU: central processing unit; ECE: expected calibration error; EEG: electroencephalography; EOG: electrooculography; ET: eye tracking; GAT: graph attention network; GPU: graphics processing unit; GRU: gated recurrent unit; IG: integrated gradients; INS-HDGS-CMT: Interpretable Neuro-Symbolic Hybrid Dynamic Graph Spiking Cross-Modal Transformer; LIF: leaky integrate-and-fire; LOSOCV: leave-one-subject-out cross-validation; LSTM: long short-term memory; Macro-F1: macro-averaged F1-score; MCC: Matthews correlation coefficient; MLP: multilayer perceptron; MMD: maximum mean discrepancy; PLV: phase-locking value; PR-AUC: area under the precision-recall curve; ROC-AUC: area under the receiver operating characteristic curve; ROI: region of interest; SD: standard deviation.

Supplementary information. The online version contains supplementary material.

Additional file 1: Figure S1. Subject-wise LOSOCV performance. **Figure S2.** Regional EEG signatures for the case-study exemplars (S24 HIGH, S30 LOW): band power by region (frontal vs. posterior), the off-diagonal functional-connectivity matrix, and the within-subject HIGH-LOW band-power difference for frontal theta and posterior alpha. **Figure S3.** ROI saliency vector (5×2 grid) of the two case-study exemplars. **Table S1.** EEG branch under both labelling regimes (robustness check). **Table S2.** Top integrated-gradients input attributions. **Table S3.** Connectivity-measure comparison ($N = 16$, reduced scale). **Table S4.** Summary of the mathematical notation used in the Methods of the main text, in order of first use. **Table S5.** Composition of the engagement label (full weighted EEG+gaze composite definition). **Table S6.** Proposed EEG branch versus each baseline on MCC and ROC-AUC (Holm-corrected Wilcoxon, 37 folds). **Figure S4.** Connectivity-threshold sensitivity (graph density and fragmentation versus τ). **Figure S5.**

Subject-count learning curve. **Figure S6.** ROI-grid resolution sensitivity. **Figure S7.** Soft-rule premises of the rule-only model projected onto electrodes and frequency bands (all eight rules). **Figure S8.** Per-subject scatter of the full model against the gaze-free branch and the tuned ET-LSTM. **Table S7.** Connectivity-threshold sensitivity (structure and downstream metrics). **Table S8.** Baselines under the earlier common fixed budget. **Table S9.** Measured cost of the spiking front-end on CPU and GPU. **Table S10.** Full model versus each eye-tracking and fusion baseline (Holm-corrected Wilcoxon). **Table S11.** ROI-grid sensitivity (saliency vector and model). **Table S12.** Label-free per-subject thresholds, transductive and causal. **Table S13.** Fidelity of the soft-rule pathway. **Table S14.** Contribution of the eye-tracking pathway. **Table S15.** Linear recoverability of the label from its own defining terms. **Table S16.** Positive control (resting state versus browsing) and behavioural purchase label. **Table S17.** Fold-wise-label robustness of the full model and the gaze-free branch.

Declarations

Availability of data and materials. The dataset analysed during the current study is the publicly available NeuMa dataset described by Georgiadis et al. [34], which is deposited in the figshare repository under a CC BY 4.0 licence as a raw release, <https://doi.org/10.6084/m9.figshare.22117001> [58], and a preprocessed release, <https://doi.org/10.6084/m9.figshare.22117124> [59]. No new datasets were generated during the current study. The code supporting the findings of this study is available at <https://github.com/priyadharshini-D29/INS-HDGS-CMT>. All experiments use a base random seed (seed = 42, Table 3) applied to the Python, NumPy, and PyTorch generators, with each fold’s prediction averaged over a 5-member ensemble; hyperparameters, optimiser configuration, and training schedule are listed in full in Table 3; software versions (Python 3.12, PyTorch 2.7.1, CUDA 12.6) are as stated in Section 2.5.10.

Competing interests. The authors declare that they have no competing interests.

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Authors' contributions. PD conceived and designed the study, developed the methodology and software, curated the data, performed the formal analysis and experiments, prepared the figures, and wrote the original draft. SS contributed to the conceptualisation, supervised the work, administered the project, acquired the resources and computing support, and reviewed and edited the manuscript. All authors read and approved the final manuscript.

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Authors' information. Not applicable.

Ethics approval and consent to participate. The NeuMa data were collected in accordance with the Declaration of Helsinki under the ethical approval reported in the original dataset publication [34]; the present study constitutes a secondary analysis.

Consent for publication. Not applicable.

Use of generative artificial intelligence. Generative artificial intelligence (AI) was used solely for language editing and grammar improvement during manuscript preparation. All content was reviewed and approved by the authors, who are fully responsible for the final manuscript.

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